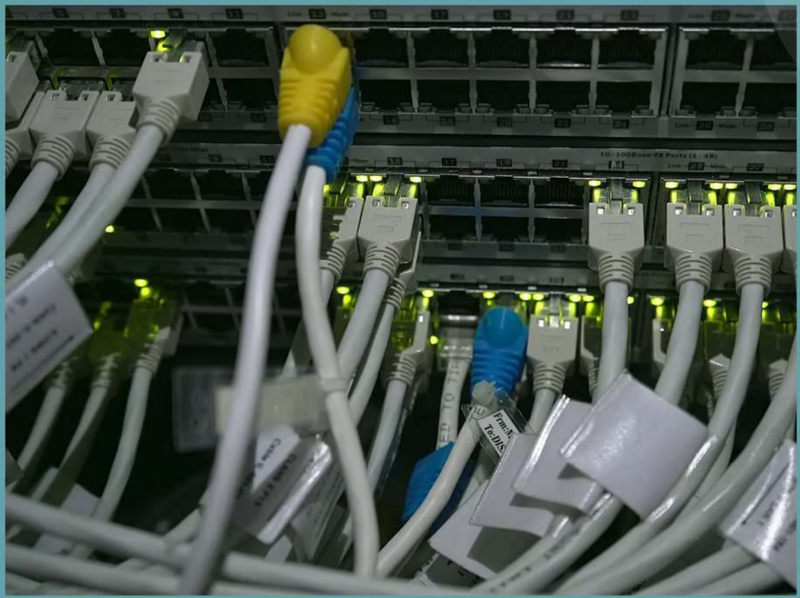
Lecture 08: Digital Communications & Networks: Part II



Information Technology Fundamentals
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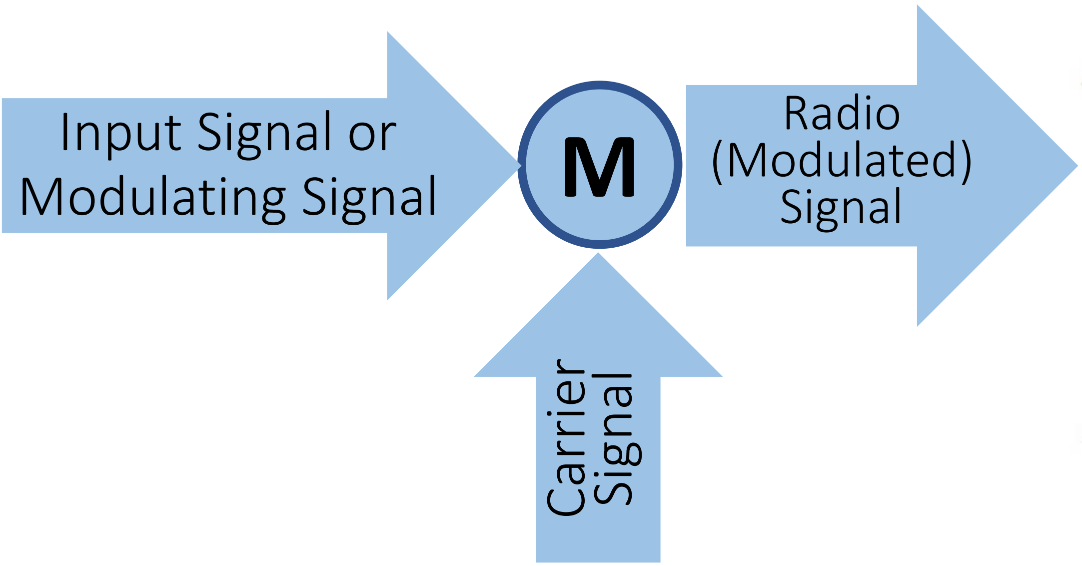
School of Information Technology
King Mongkut’s Institute of Technology Ladkrabang

Topics

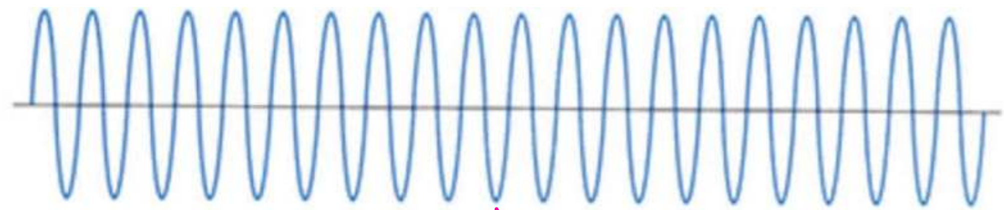
- Communication Signal Modulation (การกล้ำสัญญาณสื่อสาร)
- Multiplexing in Computer Networks (การมัลติเพล็กซ์หรือการรวมสัญญาณในเครือข่ายคอมพิวเตอร์)
- Communication Switching
- Network Models (แบบจำลองเครือข่าย)
- Protocols and Architecture

Signal Modulation (การกล้ำสัญญาณ)



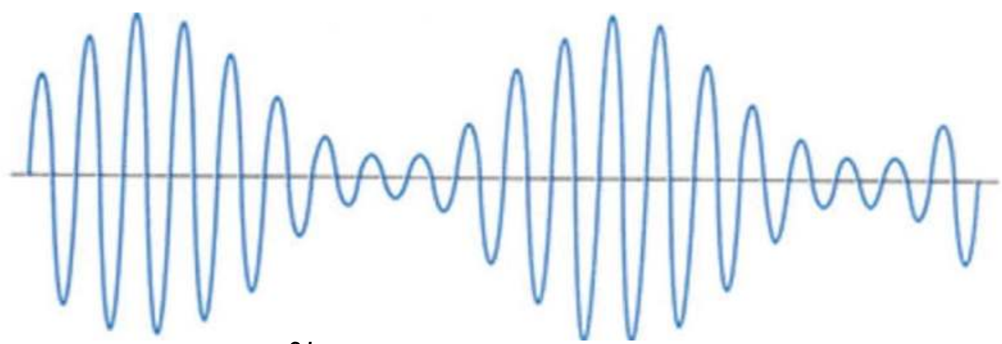
ไม่เท่า

สัญญาณข้อมูล (Data/Modulating Signal)



เท่า

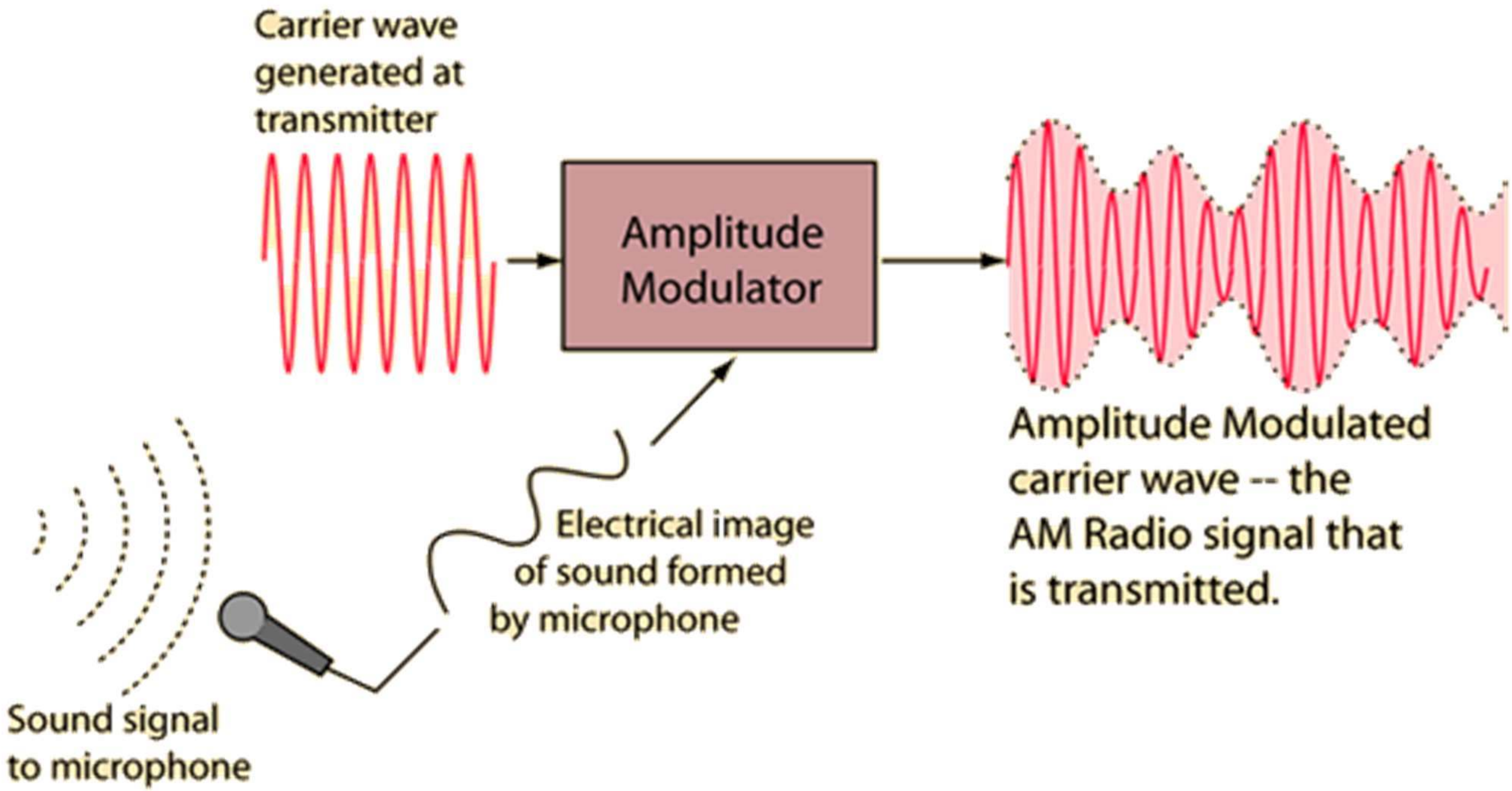
สัญญาณพาห์/คลื่นพาห์ (Carrier Signal/Wave)



สัญญาณมอดูเลต AM
(Amplitude Modulated Signal: AM Signal)

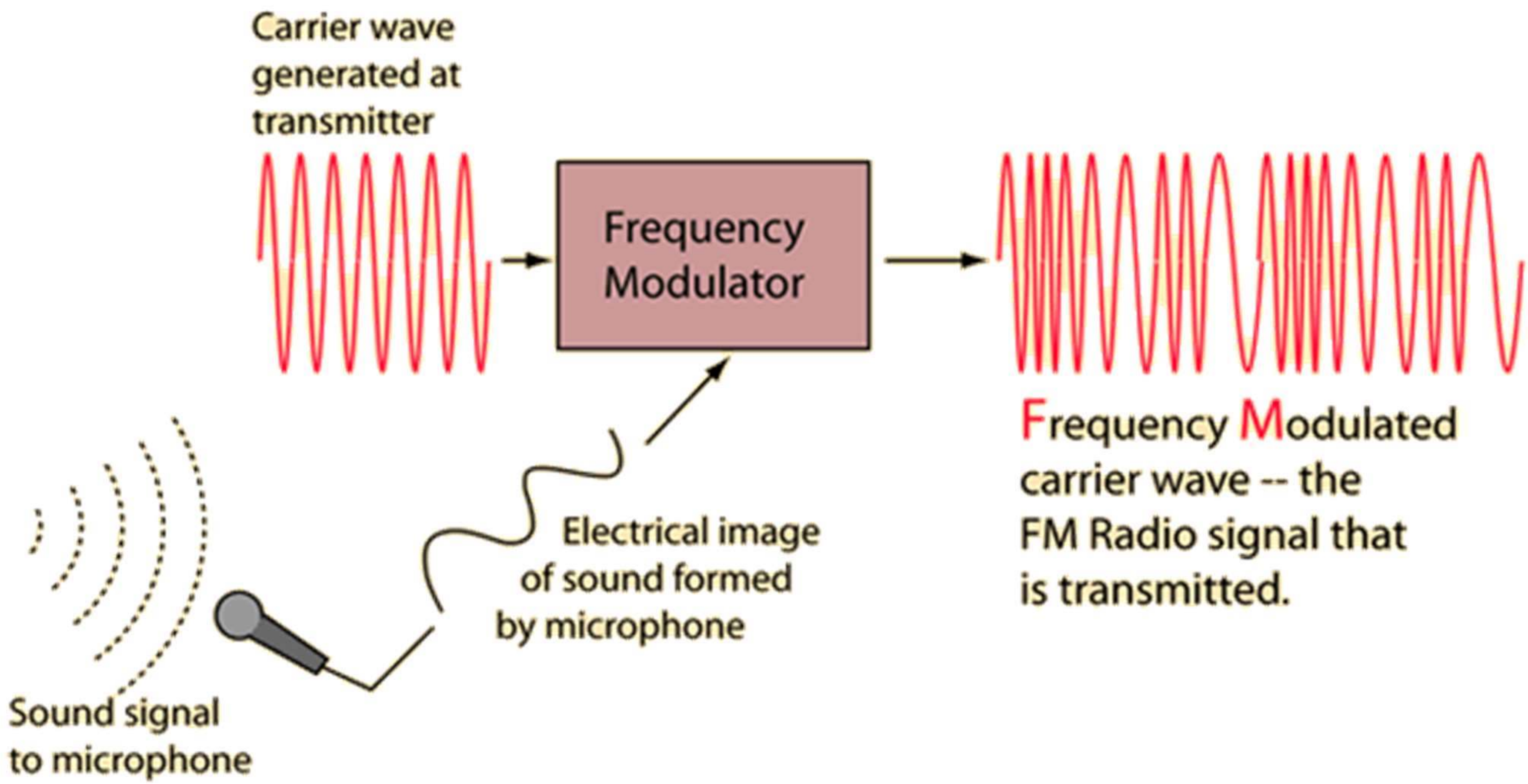
- Modulation is the process of imposing an input signal onto a carrier signal/wave.
- A carrier signal is one with a steady waveform – constant height, or amplitude, and frequency.

AM Radio

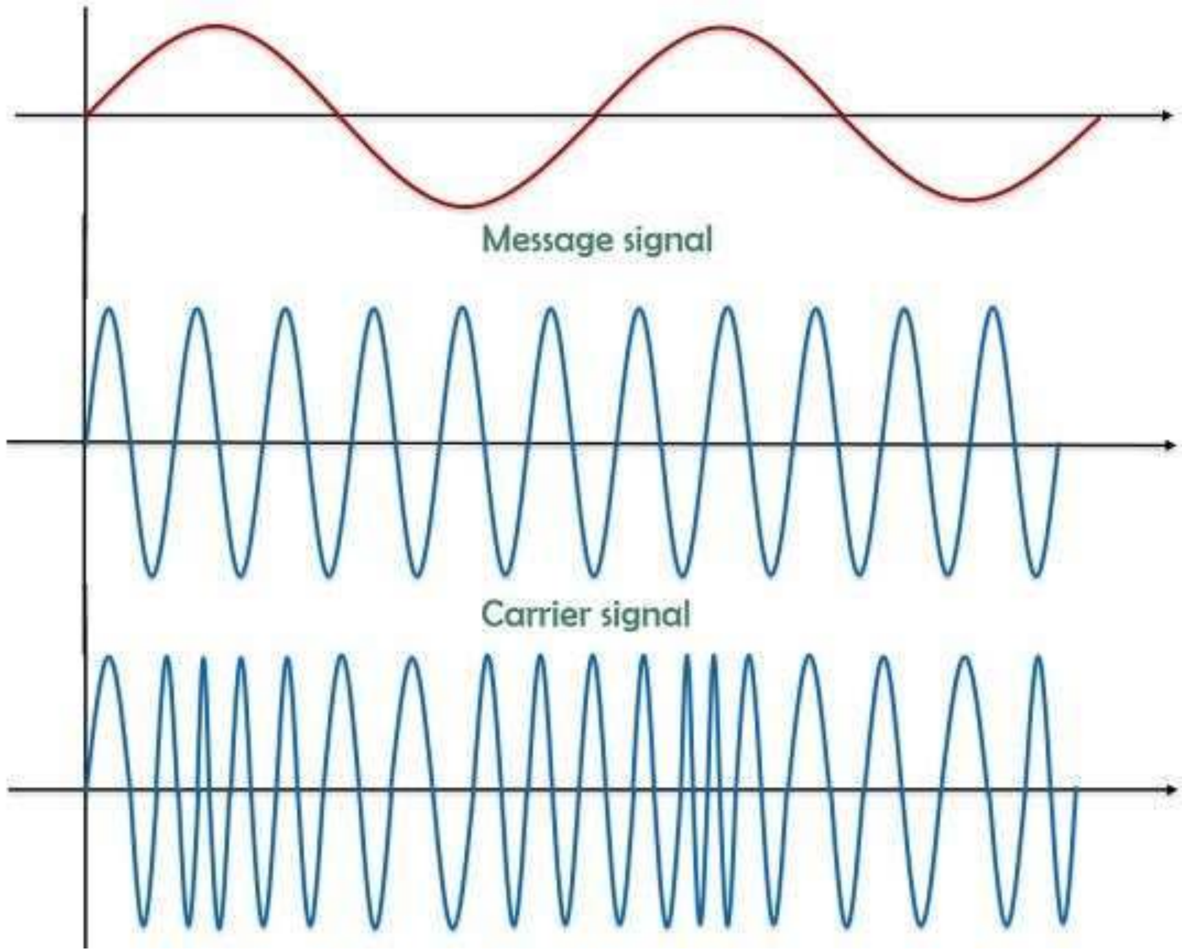


Carrier wave generated at transmitter
Amplitude Modulator
Amplitude Modulated carrier wave -- the AM Radio signal that is transmitted.
Electrical image of sound formed by microphone
Sound signal to microphone

FM Radio


Carrier wave
generated at
transmitter
Frequency
Modulator
Frequency Modulated
carrier wave -- the
FM Radio signal that
is transmitted.
Electrical image
of sound formed
by microphone
Sound signal
to microphone

Frequency Modulation (FM)
การกล้ำความถี่

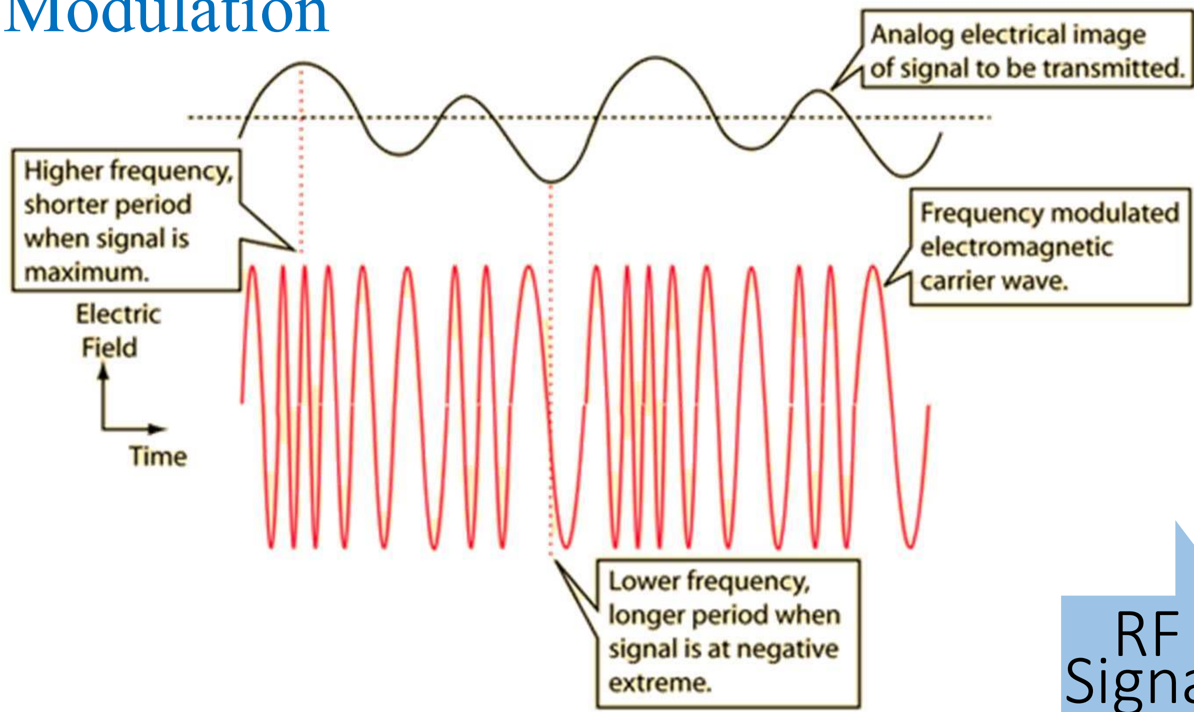


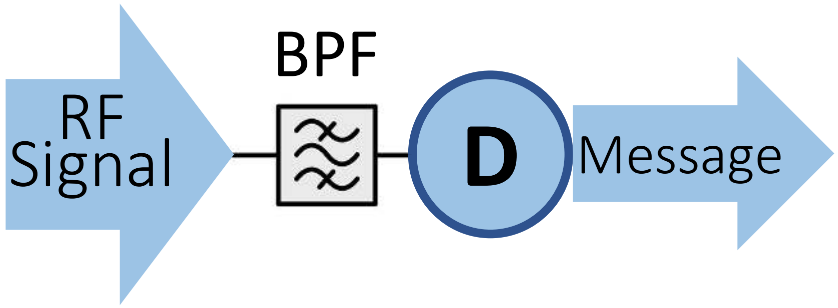
สัญญาณมอดูเลต FM (Frequency Modulated Signal: FM Signal)

Frequency Modulation (FM)

ด้านส่ง (Modulator in Transmitter)

Modulation

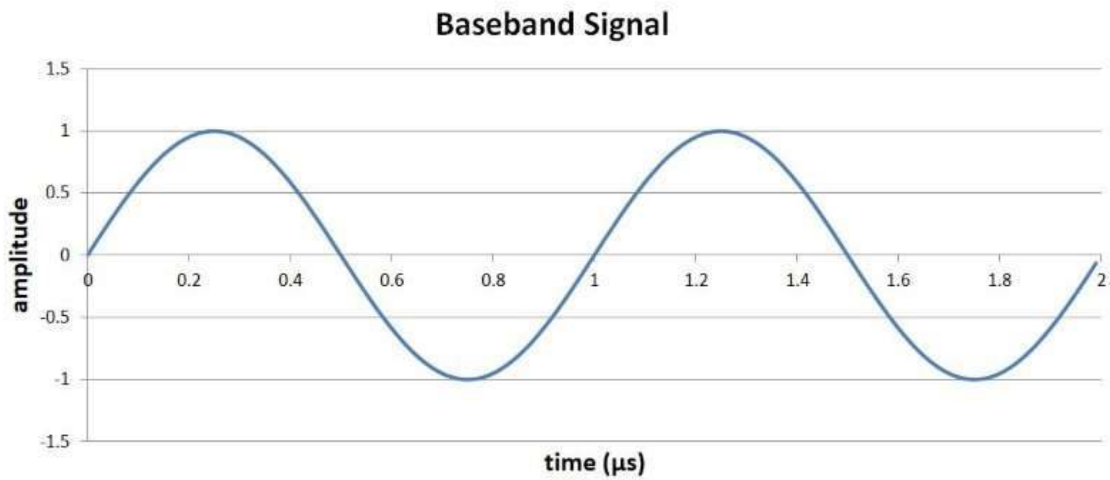




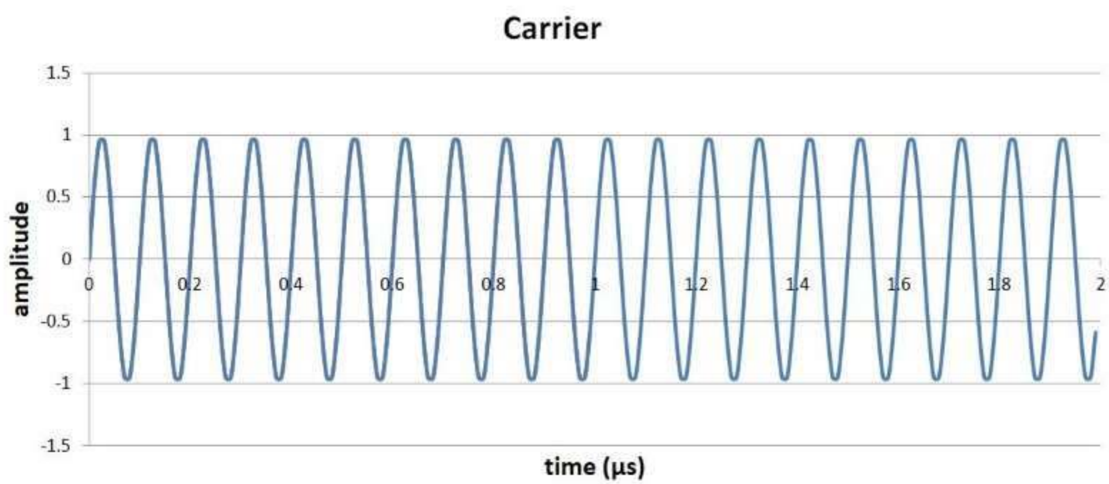
ด้านรับ (BPF & Demodulator in Receiver)

Demodulation – กระบวนการแยกสัญญาณข้อมูลจากคลื่นพาห์

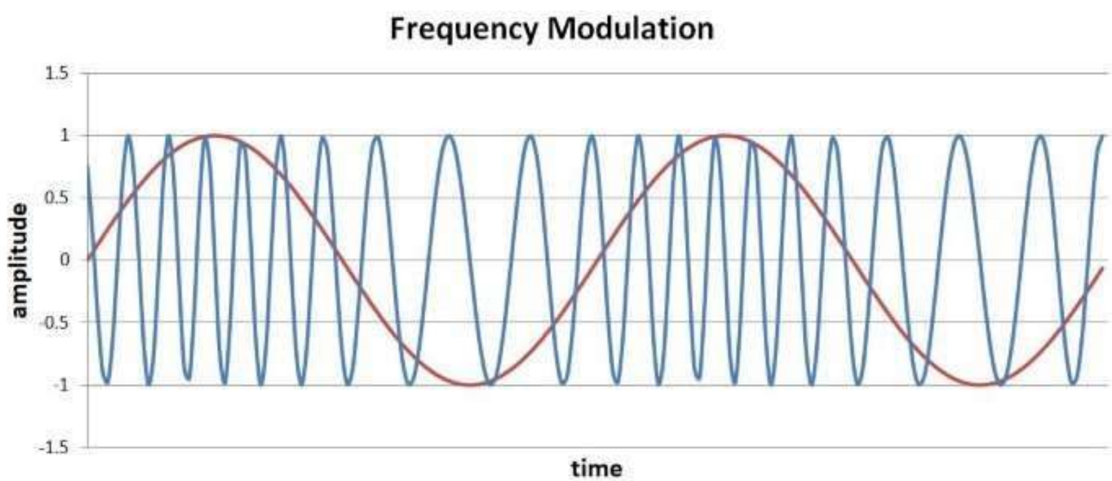
Example: FM



1 MHz Baseband Signal
(Audio Signal)

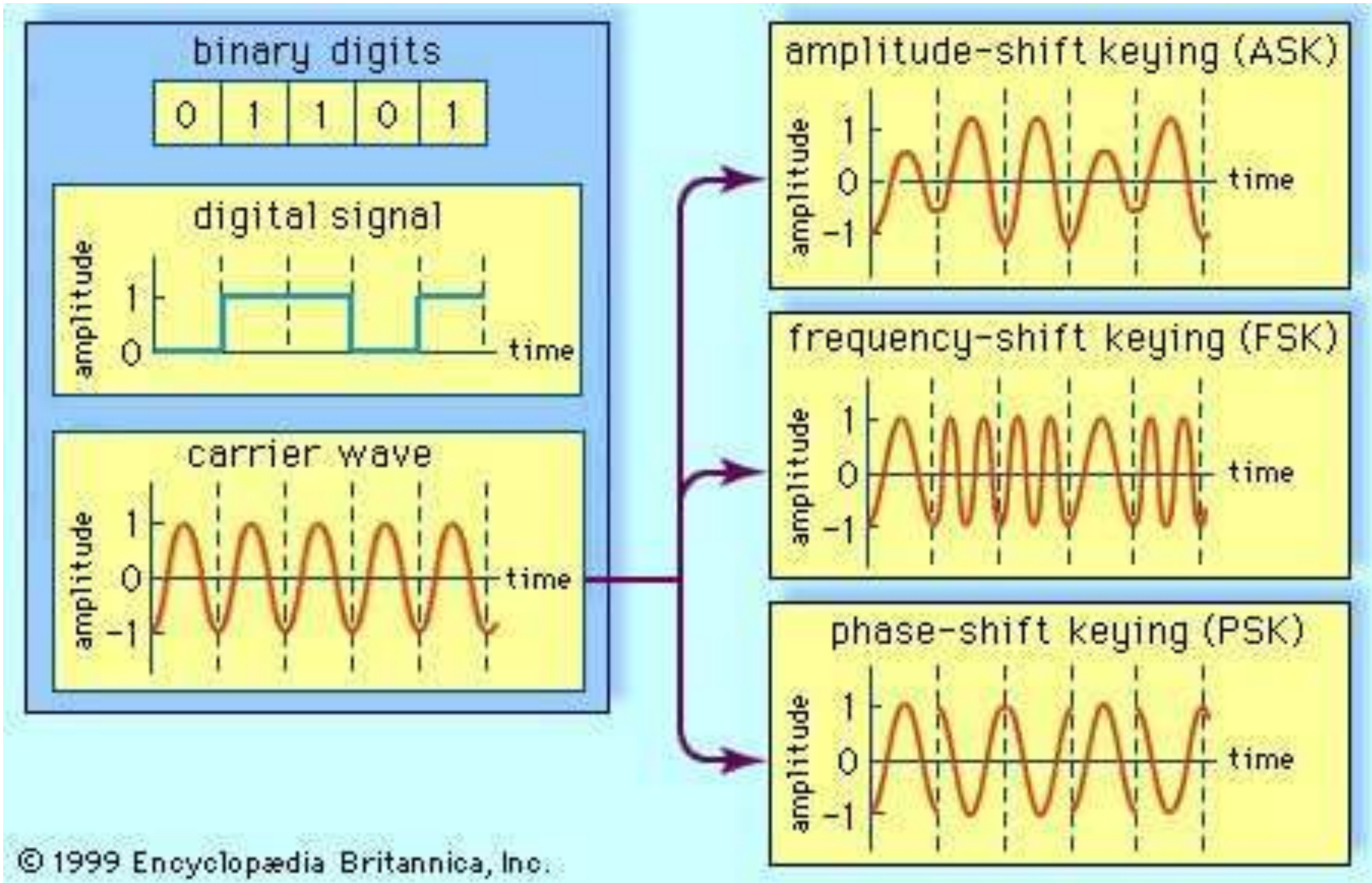


10 MHz Carrier



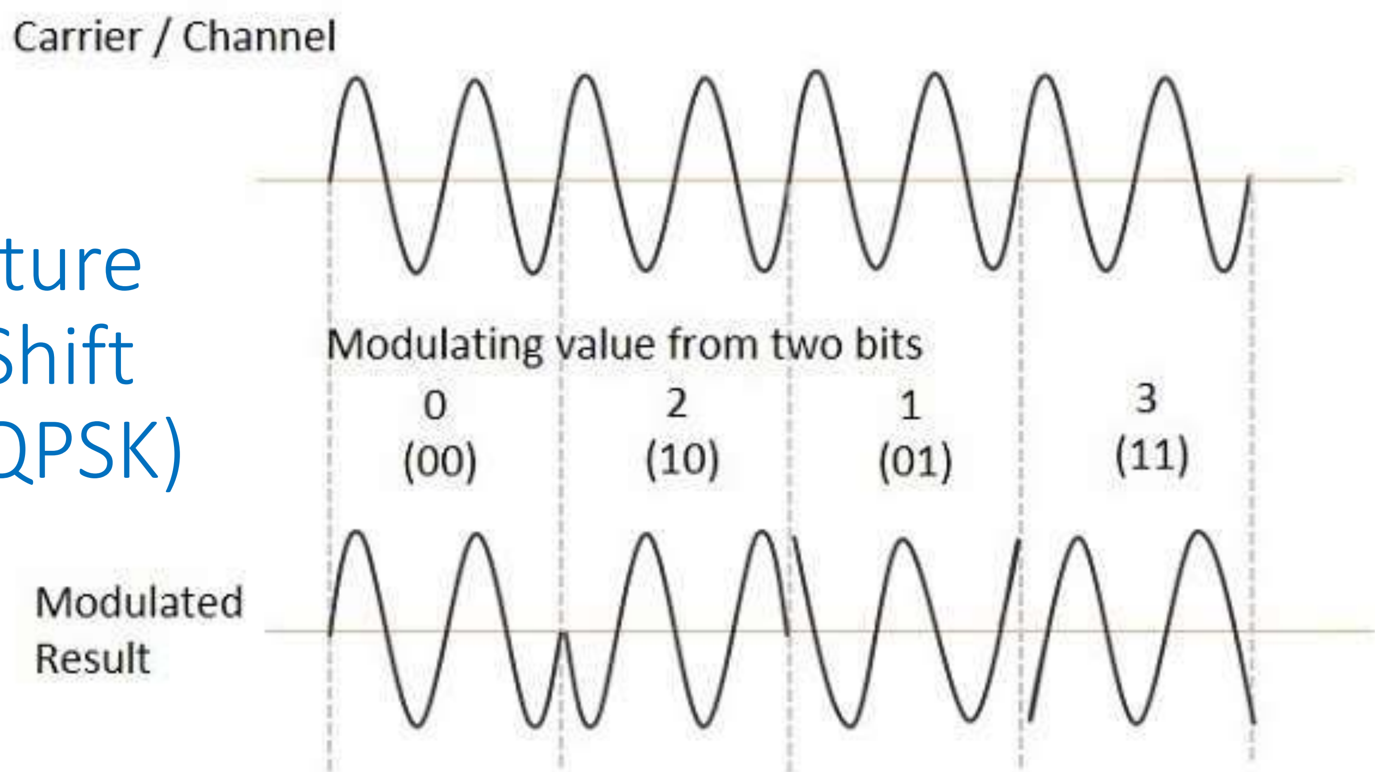
Modulated Signal
(Analog FM)

Digital Modulation Schemes – ASK (Digital AM), FSK (Digital FM) and PSK



QPSK: A Digital Modulation Method Used in 4G-LTE & 5G

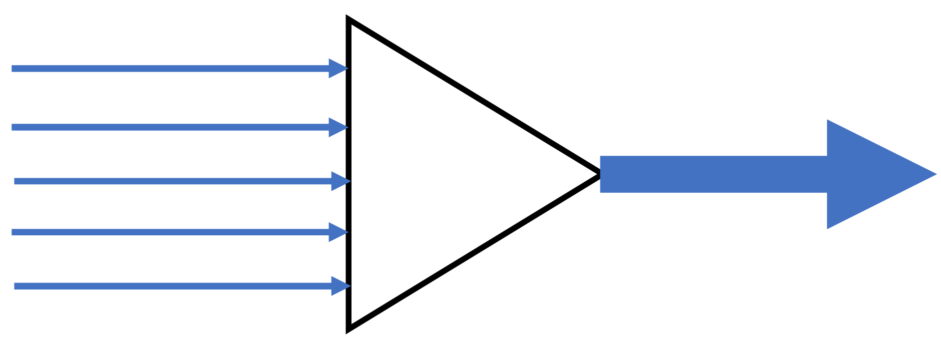
Quadrature Phase Shift Keying (QPSK)



* Modulation method in 4G LTE: QPSK, 16QAM, 64QAM.
* Modulation method in 5G: π/2-BPSK, QPSK, 16QAM, 64QAM, and 256QAM

Multiplexing in Computer Networks
การรวมสัญญาณเพื่อการแบ่งปันช่องทางสื่อสาร

- Whenever the bandwidth of a medium linking two devices is greater than the bandwidth needs of the devices, the link can be shared.
- Multiplexing is the set of techniques that allows the simultaneous transmission of multiple signals across a single data link.



Multiplexing in Computer Networks
การรวมสัญญาณเพื่อการแบ่งปันช่องทางสื่อสาร

Frequency Division Multiplexing (FDM)

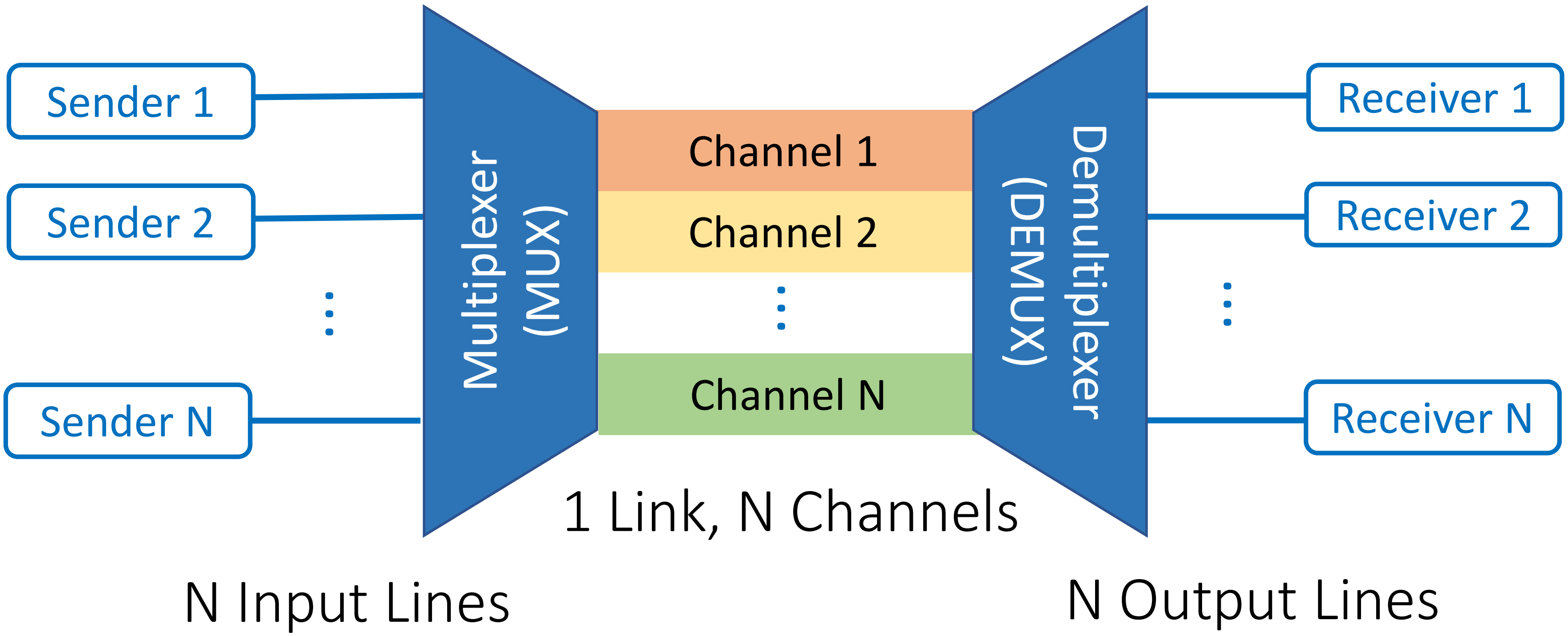
Time Division Multiplexing (TDM)

Wavelength Division Multiplexing (WDM)

Code Division Multiplexing (CDM)

Orthogonal Frequency Division Multiplexing (OFDM)

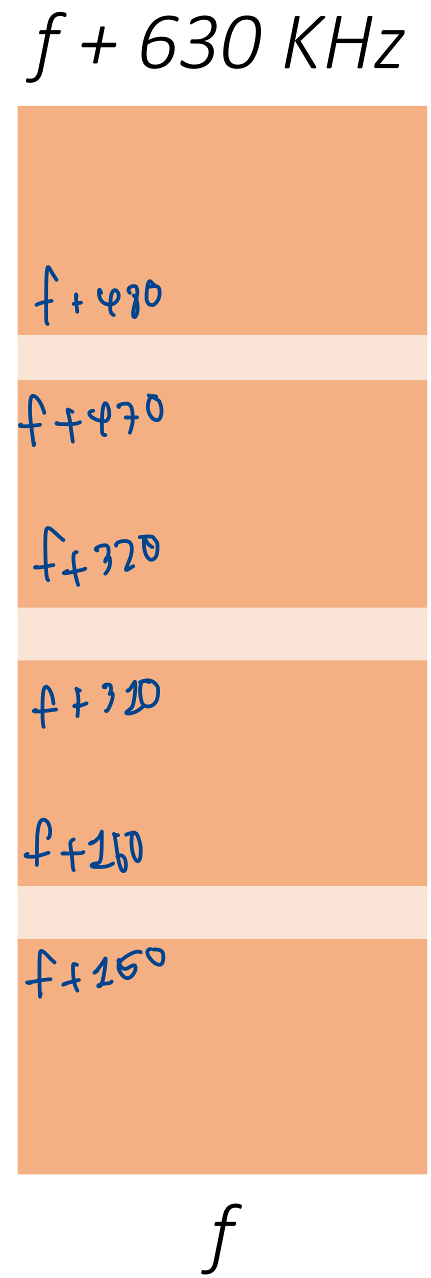
Frequency Division Multiplexing (FDM) การรวมสัญญาณแบบแบ่งความถี่



FDM is the concept of dividing a communication channel to several non-overlapping channels (a set of independent channels) with different bandwidths.

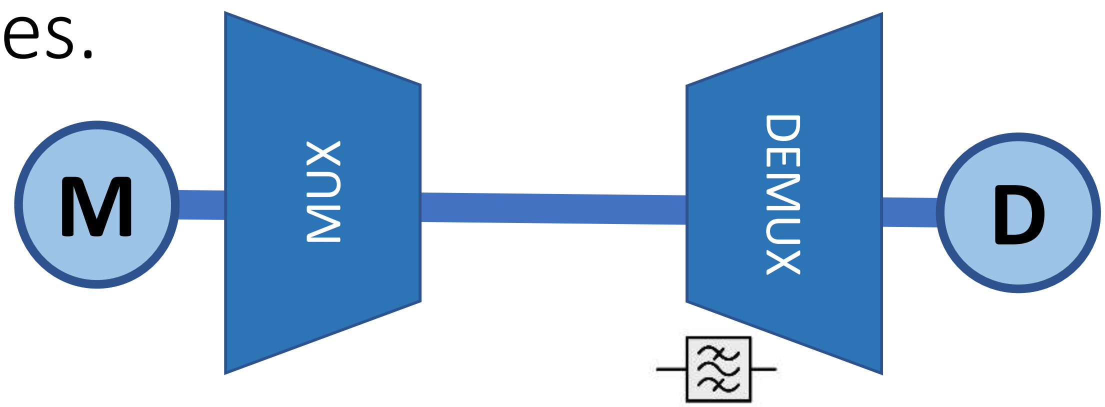
Example of FDM

- Consider four frequency bands (N=4), each with a known finite bandwidth of 150 kHz and separated by three guard bands (แถบความถี่ป้องกัน) of 10 kHz each.
- To accommodate all the bands, the communication channel should have a capacity of (150 x 4) + (10 x 3) = 630 KHz.
- FDM multiplexes the four frequency bands and sends them via the communication channel (1 link, 4 channels).

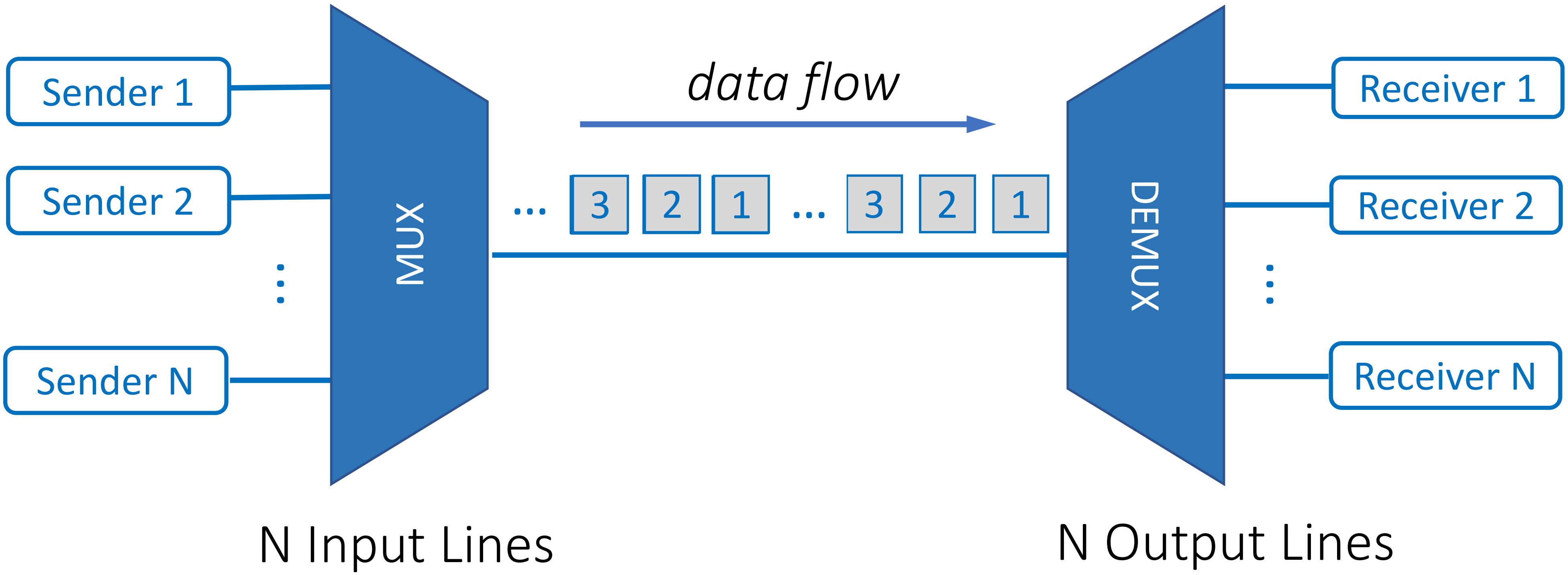


Example of FDM (cont.)

- Each message modulates a different carrier, so the modulated signals are in different frequency bands that don't interfere with each other.
- It can use any AM or FM technique.
- The signal is then demultiplexed at the receiver's end with a bank of bandpass filters and demodulators to regenerate the original frequencies.

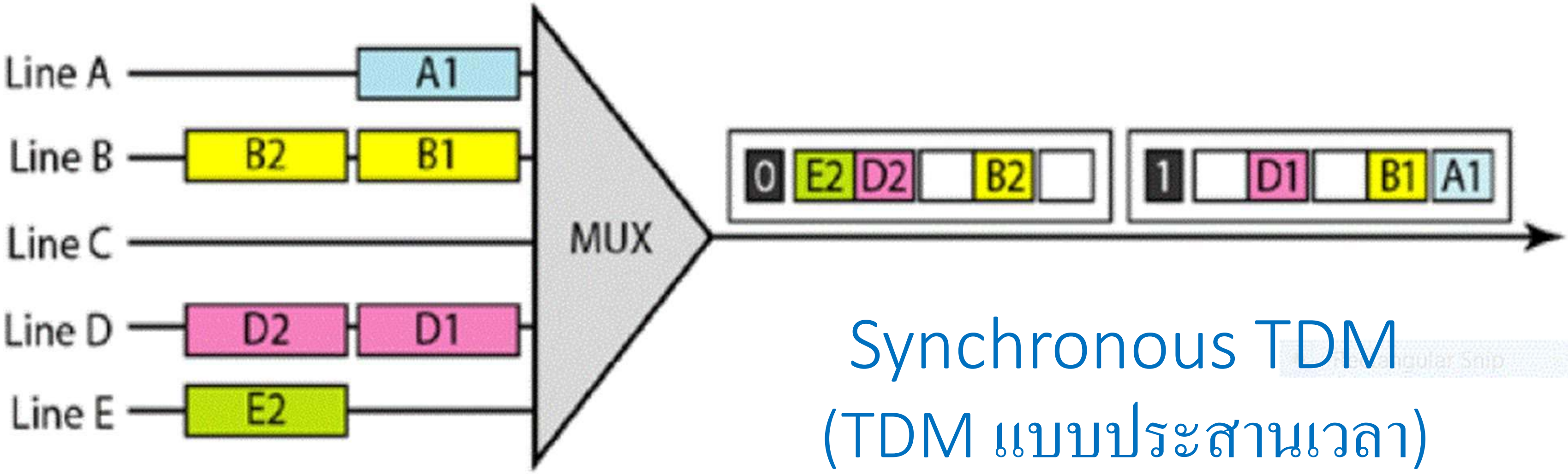


Time Division Multiplexing (TDM)
การรวมสัญญาณแบบแบ่งเวลา

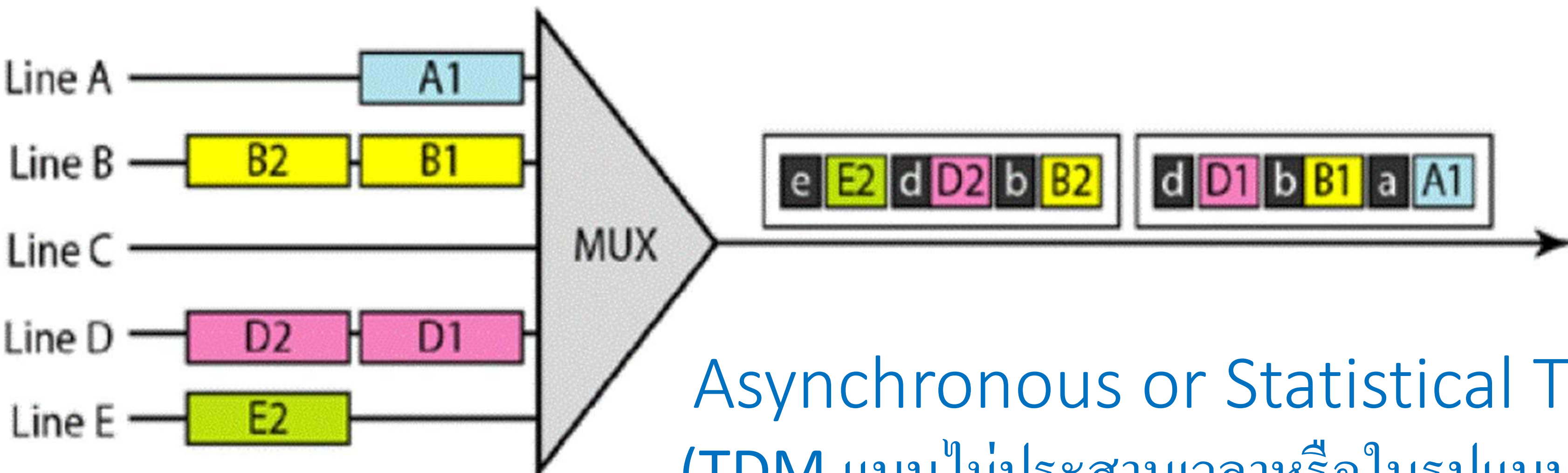


Time Division Multiplexing (TDM) Conceptual View
with items from multiple sources sent over a shared medium

Time Division Multiplexing (TDM): Synchronous vs Statistical TDM

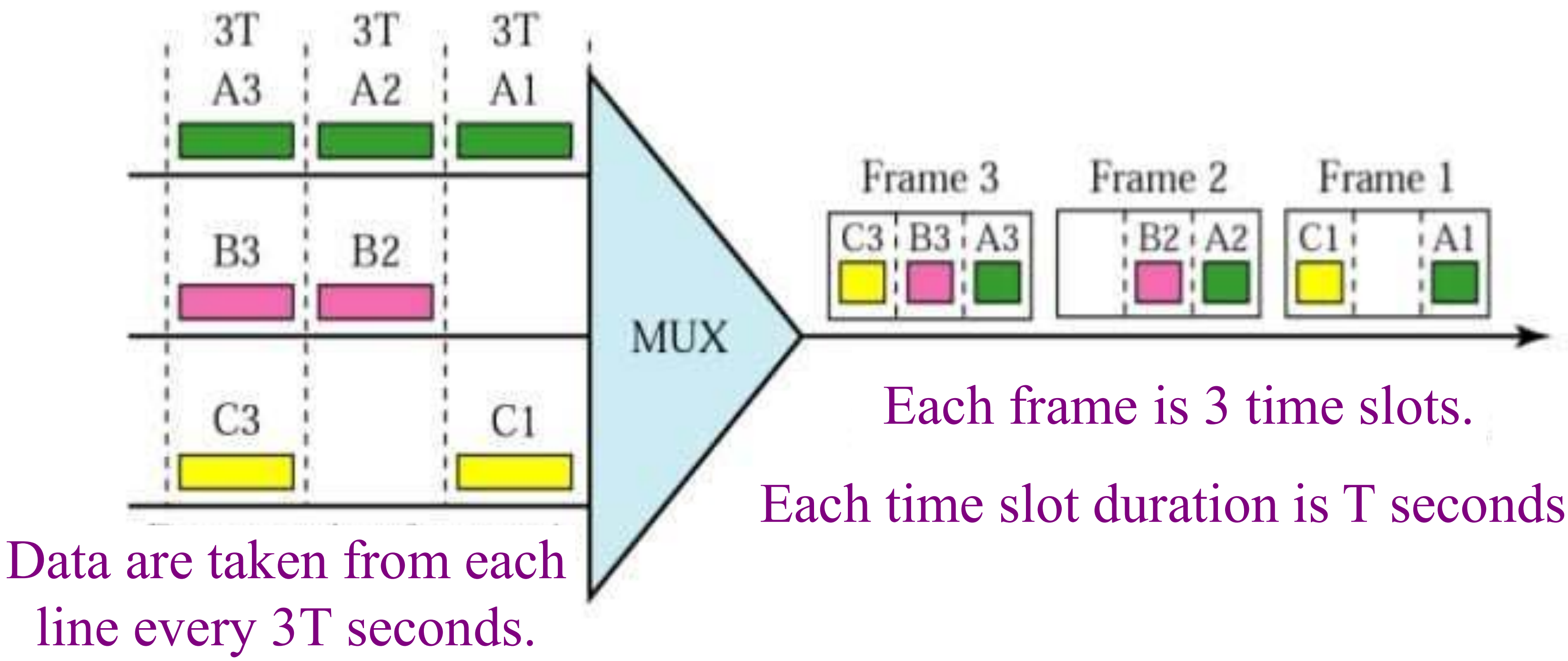


Synchronous TDM
(TDM แบบประสานเวลา)



Asynchronous or Statistical TDM
(**TDM** แบบไม่ประสานเวลาหรือในรูปแบบสถิติ)

Interleaving in Synchronous TDM
การแทรกสลับหรือการสลับ



Time Division Multiplexing (TDM): Synchronous vs Statistical TDM

- Synchronous TDM does not guarantee that the full capacity of a link in used.
- Like synchronous TDM, asynchronous TDM allows a number of lower speed input lines to be multiplexed to a single higher speed line.
- Unlike synchronous TDM however in asynchronous TDM, the total speed of the input can be greater than the capacity of the path.

Applications of TDM

1. Digital audio mixing system.
2. Half-duplex communication system.
3. Integrated Service Digital Network (ISDN) telephone system.
4. Public Switched Telephone Network (PSTN) system.
5. Synchronous Digital Hierarchy (SDH) system – a CCITT standard for a hierarchy of optical transmission rates
6. GSM communication technology.
7. Satellite Access system.
8. Cellular Radio.

Main Advantages of TDM

1. The hardware required for TDM is very less.
2. TDM circuits are not so complex than other systems like FDM. It is a great advantage of TDM over FDM.
3. In the TDM system, the full bandwidth of a channel can be used.
4. TDM provides more throughput than FDM.
5. TDM system does not need any carrier wave or carrier signal, but FDM need a carrier signal which divides the main signal according to frequency.

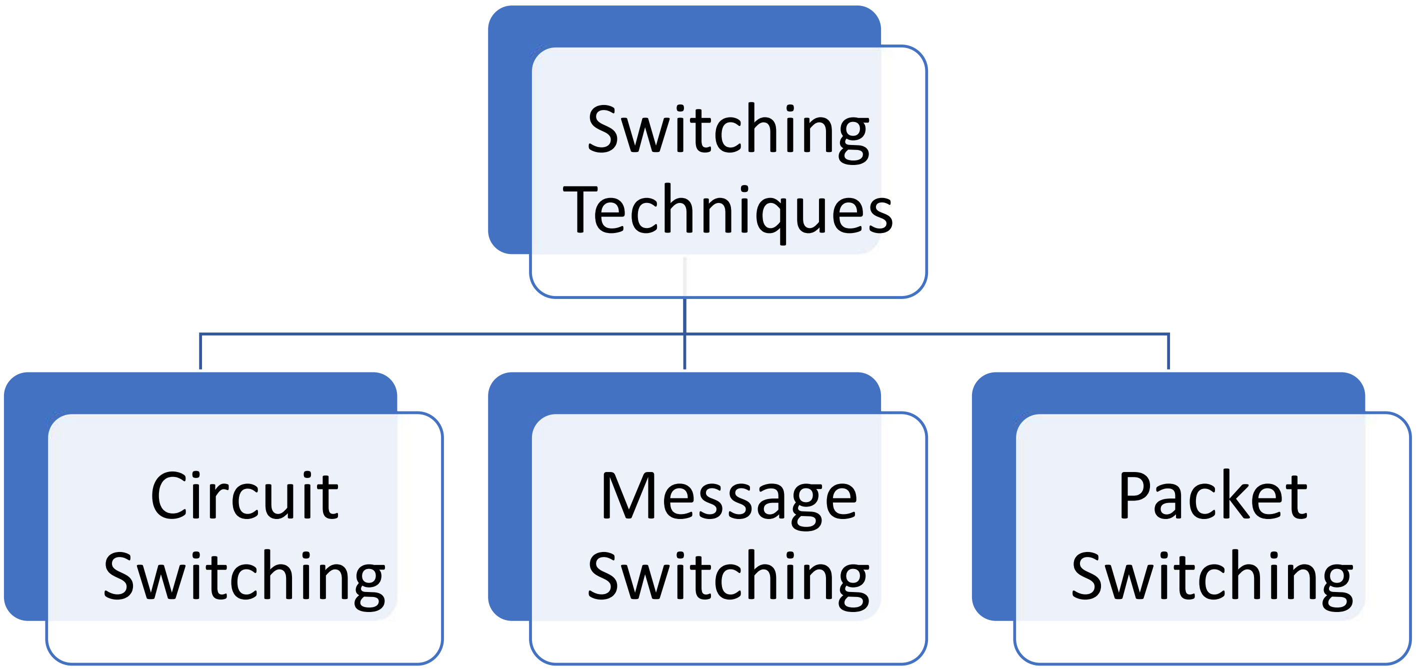
Advantages of Multiplexing in Wireless Communication Systems

- Reducing the communication cost.
- Enhancing the connectivity.
- Saving valuable communication resources.
- Improving network capacity.
- Eliminating the need of exclusive links between sources and destinations.

Communication Switching

- The switching function provides communication pathways between two endpoints and manages how data flows between them.
- In large networks, there can be multiple paths from sender to receiver. The switching technique will decide the best route for data transmission.
- Switching technique is used to connect the systems for making one-to-one communication.

Communication Switching Methods



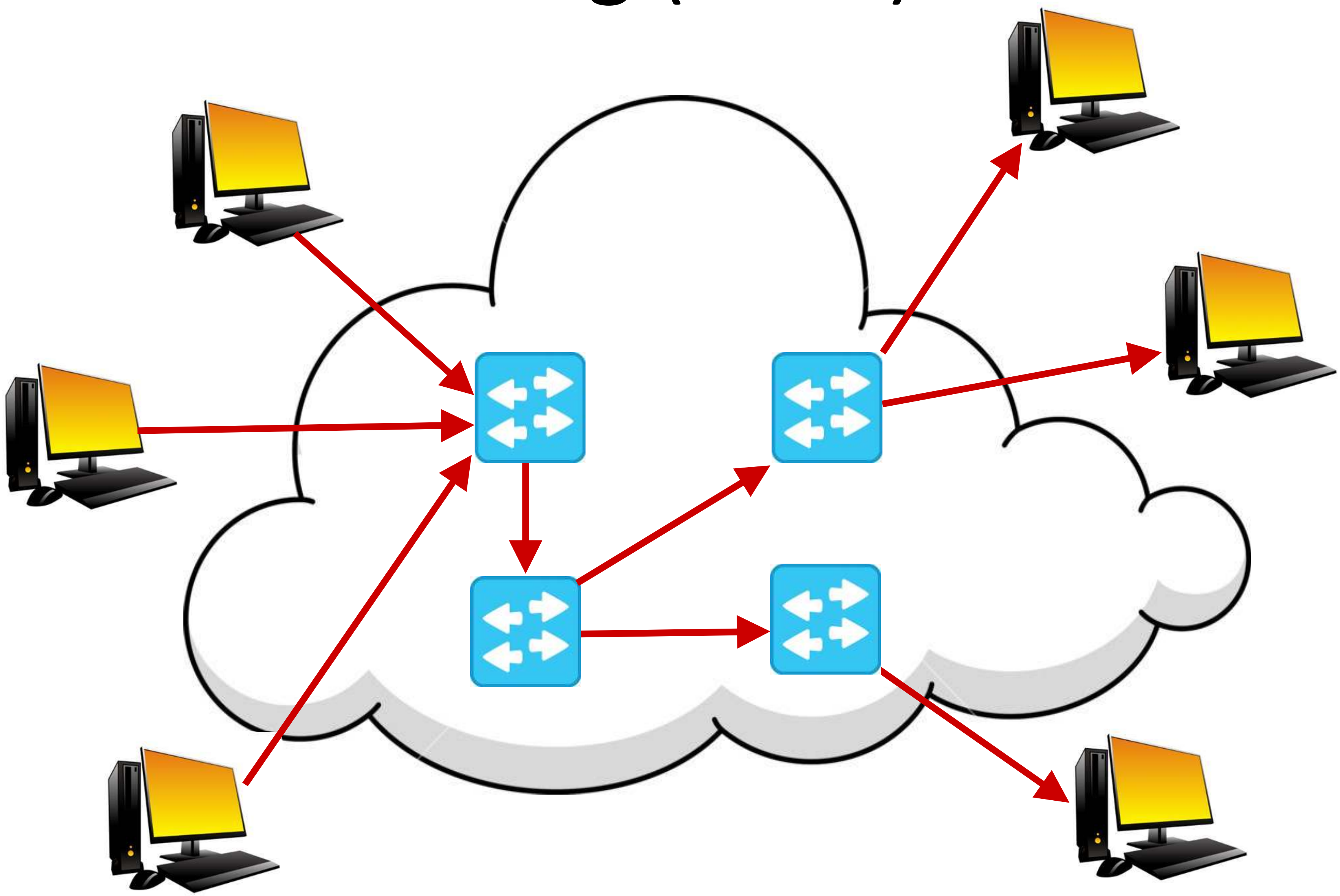
เซอร์กิตสวิตชิง
(การสลับวงจร)

แพ็กเก็ตสวิตชิง
(การสลับกลุ่มข้อมูล)

Circuit Switching

- It is a switching method in which a dedicated physical path is formed between two points in a network between the sending and the receiving devices.
- These dedicated paths are created by a set of switches connected by physical links.
- Circuit switching is the simplest method of data communication that has a fixed data rate and both the subscribers need to operate at this fixed rate.

Circuit Switching (cont.)



Circuit-Switched Network (CSN)

- A network based on circuit switching
- A circuit-switched network (CSN) is a type of network where the communications between end devices (nodes) must be set up before they can communicate.
- Once set up, the circuit is dedicated to the two nodes it connects for the duration of that connection.
- An example of a circuit-switched network is an analog telephone network and ISDN.

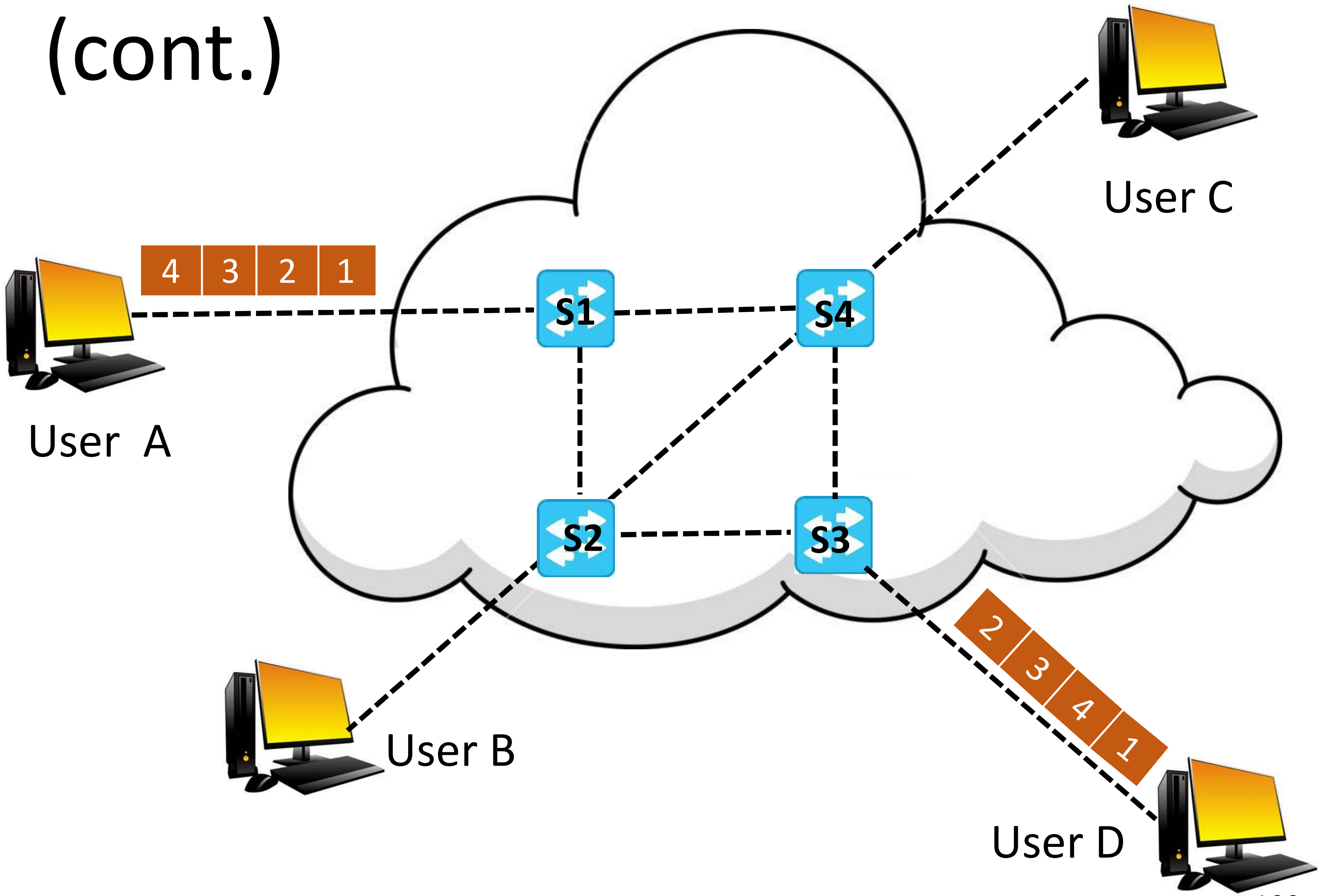
Packet Switching

- is the primary basis for data communications in computer networks
- uses one link at a time
- sends whenever the next link is free – interleave messages
- is used in the Internet and most local area networks.
- uses Store and Forward technique – each hop will store the packet and then forward the packets to the next host destination.

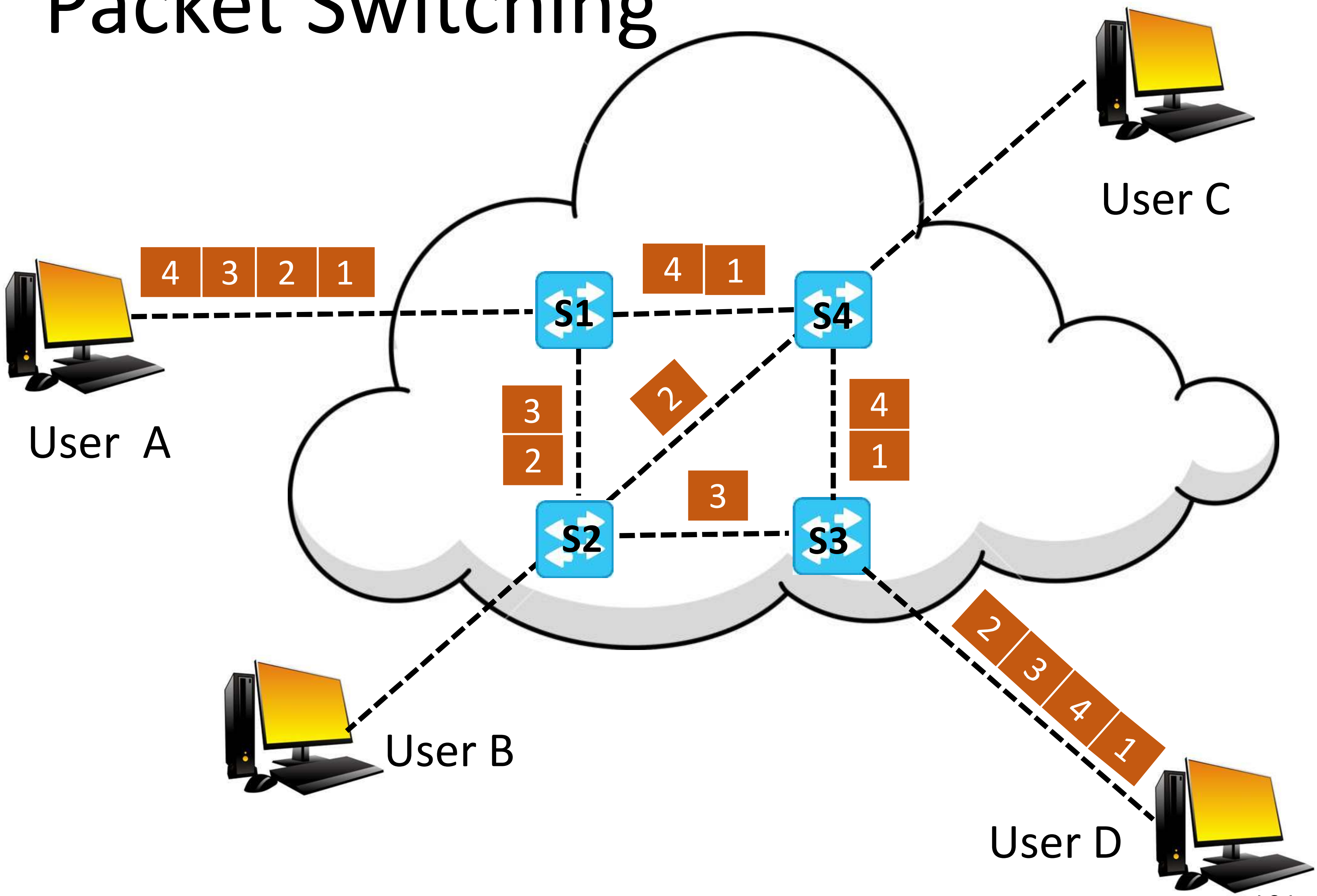
Packet Switching (cont.)
Steps

1. Data is split into packets.
2. Each packet has a source address, a destination address, and payload.
3. If multiple packets are required, then the order of each packet is noted.
4. Packets sent onto the network, moving from router to router taking different paths. Thus, each packet's journey time can differ.
5. Once packets arrive, they are reordered and reassembled.
6. Message sent from Recipient to Sender indicating that the message has been received.
7. If no confirmation message, sender transmits the data again.

Packet Switching (cont.)



Packet Switching

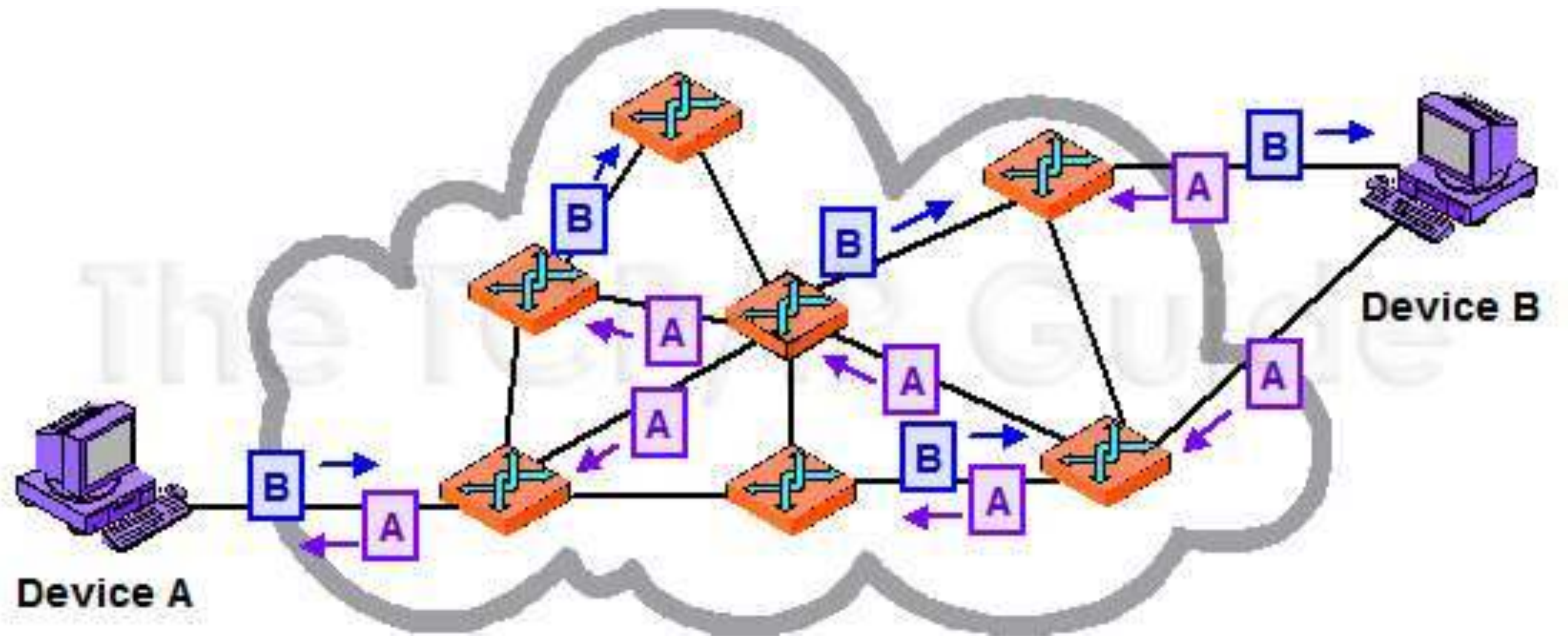


Packet-Switched Network (PSN)

- A network based on packet switching
- A packet-switched network (PSN) is a type of computer communications network that groups and sends data in the form of small packets.
- It enables the sending of data or network packets between a source and destination node over a network channel that is shared between multiple users and/or applications.
- The packets usually contain a header for addressing and error control information.
- The network layer in the Internet today is a PSN.

Packet-Switched Network (PSN)

Full Duplex Transmission



Two Types of Packet Switching

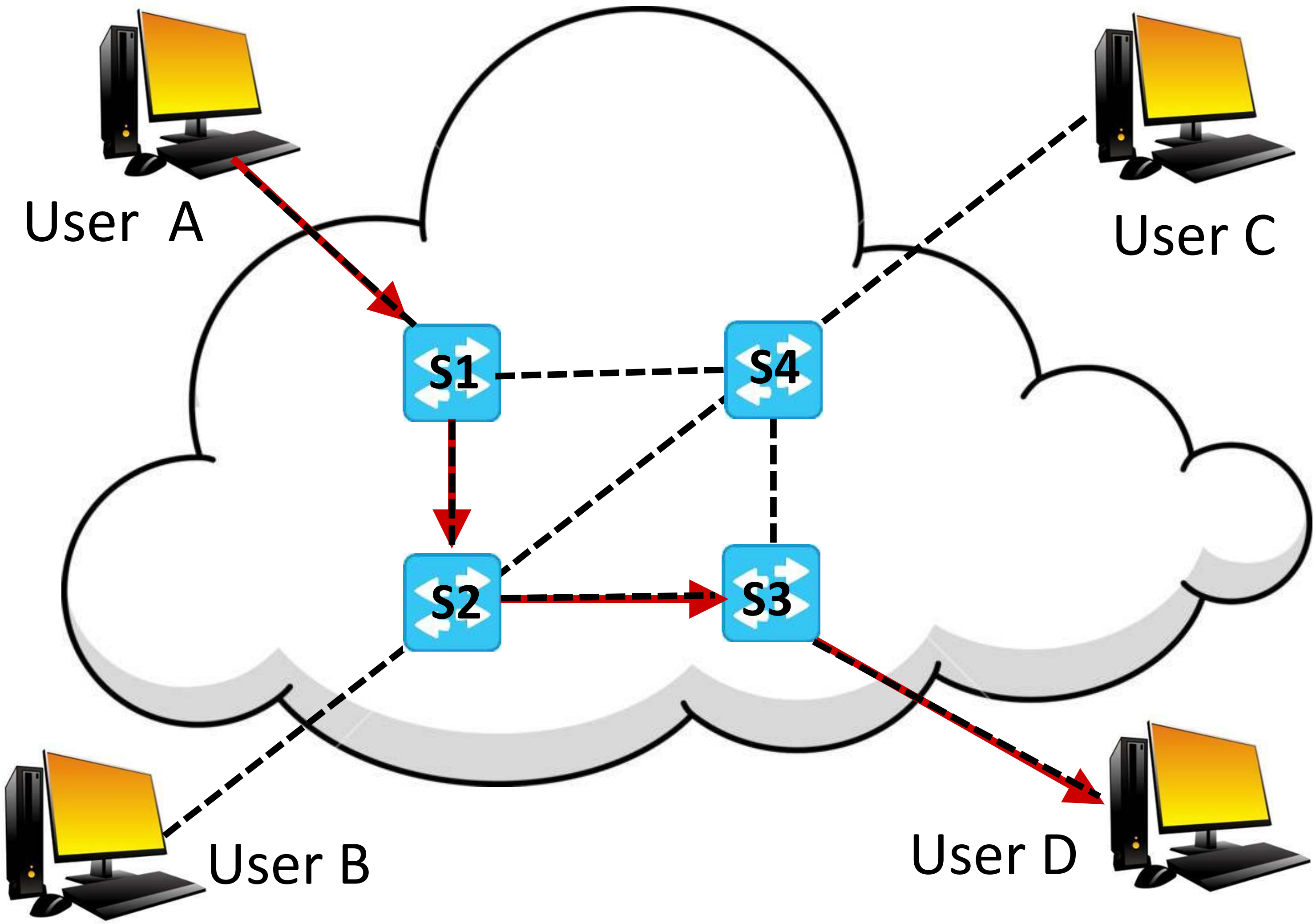
- Connection-Oriented Packet Switching (Virtual Circuit Switching: VCS)
- Connectionless Packet Switching (Datagram Switching)

Virtual Circuit approach	Datagram approach
Intermediate nodes do not take any routing decision.	Intermediate nodes take routing decisions to forward the packets.
Congestion can occur when the intermediate node is busy, and it does not allow other packets to pass through.	Congestion cannot occur as all the packets travel in different directions.
It is not very flexible.	It is more flexible as all the packets are treated as an independent entity.

Connection-Oriented Packet Switching (Virtual Circuit Switching: VCS)

- Virtual circuit switching is a hybrid technology combining features of both circuit switching and packet switching.
- A preplanned route is established before packet transmission can begin – call request and call accept packets are used to establish the connection between sender and receiver.
- The path is fixed for the duration of a logical connection.
- Virtual Circuit Identifier (VCID or VCI) is used instead of an address in each header.
- The logical path is called virtual circuit (VC) because the circuit is not real and dedicated to a user pair.

Connection-Oriented Packet Switching (Virtual Circuit Switching: VCS)



Advantages of VCS

- **Packets are delivered in order** – all packets take the same route
- **The overhead in the packets is smaller** – there is no need for each packet to contain the full address
- **The connection is more reliable** – network resources are allocated at call setup so that even during times of congestion, provided that a call has been setup, the subsequent packets should get through
- **Billing is easier** – billing records need only be generated per call and not per packet.

Connectionless Packet Switching

- No connection is established between the sender and the receiver.
- Ethernet and Internet Protocol (IP) are the two dominant technologies.
- Each packet, in its header, contains complete addressing information.
- For this reason, it is possible for the various packets to be conveyed via different routes.

Circuit Switching	Packet Switching
Done at physical layer.	Done at network layer.
Direct physical connection between sender and receiver.	No direct physical connection between sender and receiver.
Since pre-defined path is there, there are 3 different stages: connection establishment, data transfer, connection termination.	No specific path for transferring data so only data transfer takes place directly.
The entire message goes at once. So no header (extra information to recognize) is required.	Since the entire message gets segmented into different pieces, headers are required to identify at destination side.
Data will go as it is in same order.	Different parts of data reached to destination. So there is a chance that out of order will occur.
Data processed only once at source/sender side.	At each intermediate network, data will be processed including source.
If data is big, circuit switching is useful.	If data is small, packet switching is useful.
Guarantee that the entire message will be received in the same order as they were sent. So more reliable communication.	There is possibility that one of data packet may lose. So not reliable communication.
Requires more resources even for less communication.	Requires less resources.
Main disadvantage is if at any place connection is lost, entire communication must be restarted.	The main advantage over circuit switching is, if one path is lost only the packets which are on that only will be restarted. But not entire data packets.

Network Models

- The Open Systems Interconnection Model (OSI Model)
- The Transmission Control Protocol/Internet Protocol Model (TCP/IP Model)

OSI model is a generic model that is based upon functionalities of each layer.

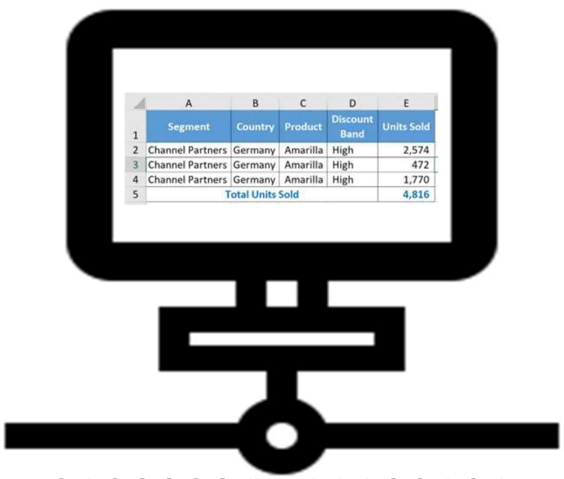
TCP/IP model is a protocol-oriented standard.

OSI model gives guidelines on how communication needs to be done, while TCP/IP protocols layout standards on which the Internet was developed. So, TCP/IP is a more practical model.

OSI Model (แบบจำลองโอเอสไอ)

7 Layers of the OSI Model

"<u>A</u>ll <u>P</u>eople <u>S</u>eem <u>T</u>o <u>N</u>eed <u>D</u>ata <u>P</u>rocessing"



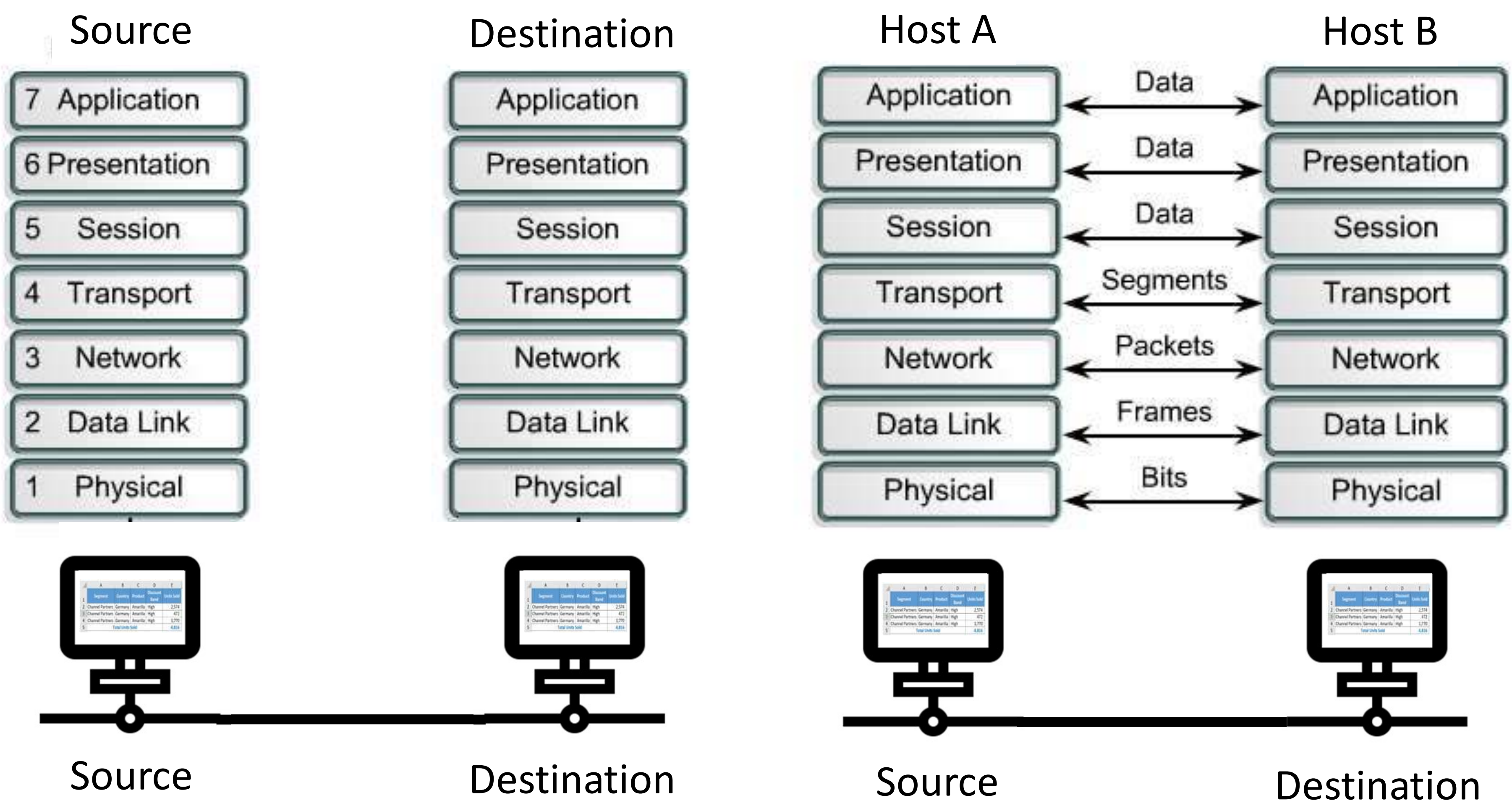
01000001...11100101

	Data Unit	Layers	#
Host Layers	Data	Application	7
	Data	Presentation	6
	Data	Session	5
	Segments	Transport	4
Media Layers	Packets	Network	3
	Frames	Data Link	2
	Bits	Physical	1
	Data Unit	Layers	#

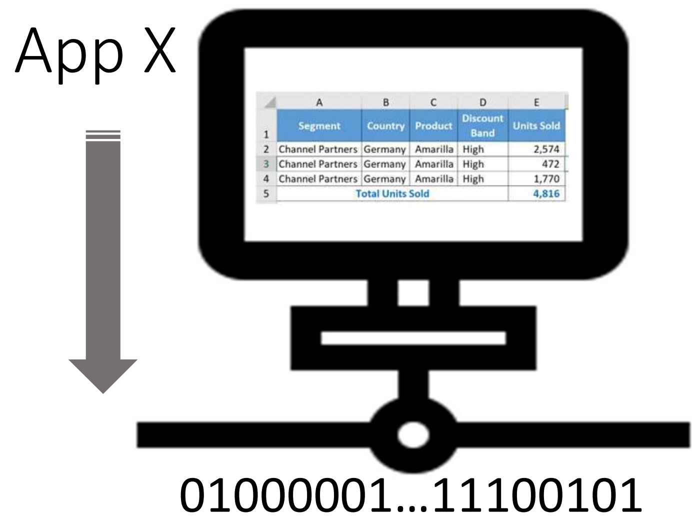
OSI Model (แบบจำลองโอเอสไอ)
7 Layers of the OSI Model (cont.)

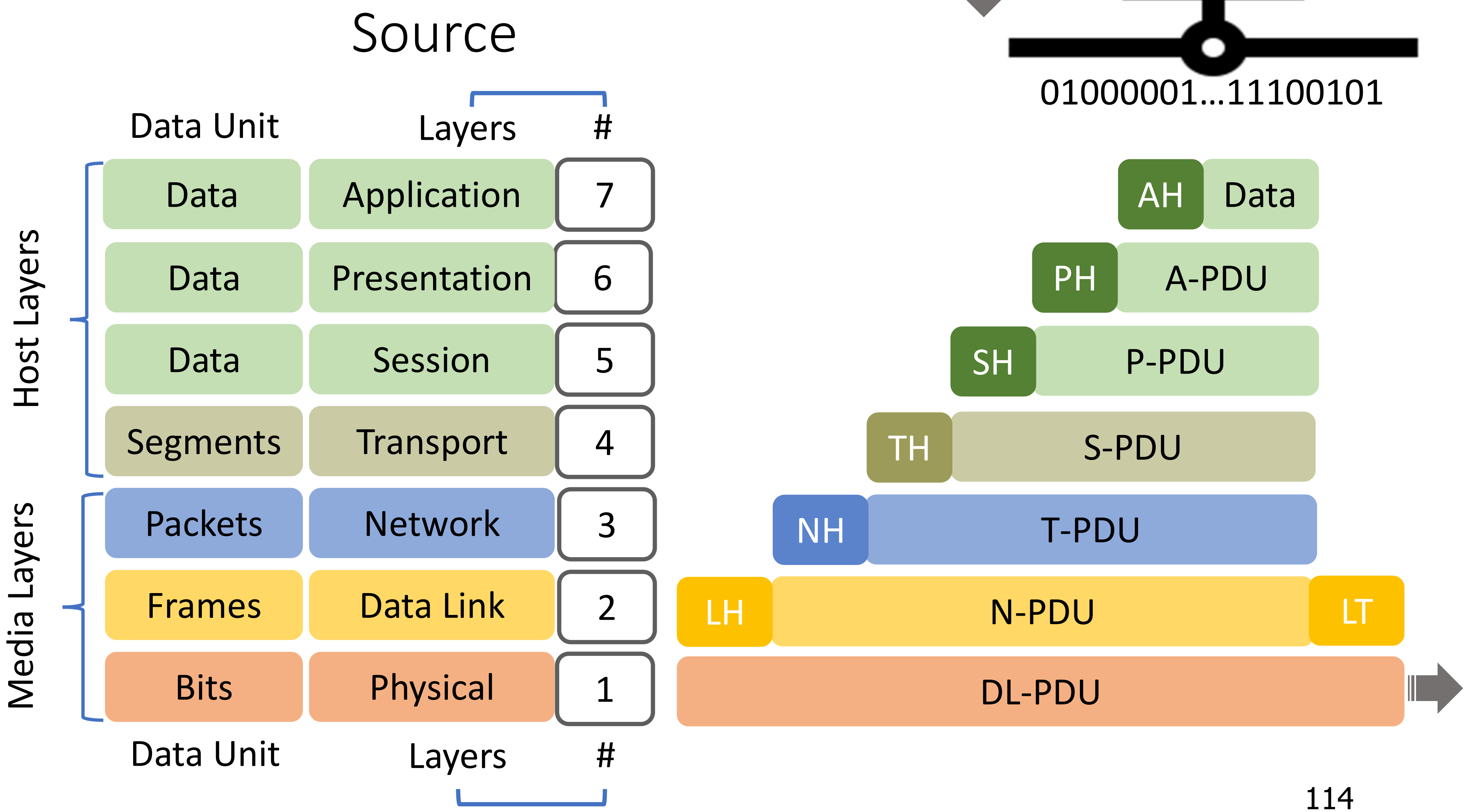
- Each layer performs a subset of the required communication functions
- Each layer relies on the next lower layer to perform more primitive functions
- Each layer provides services to the next higher layer
- Changes in one layer should not require changes in other layers

OSI Model (cont.)
Peer-to-Peer Communication



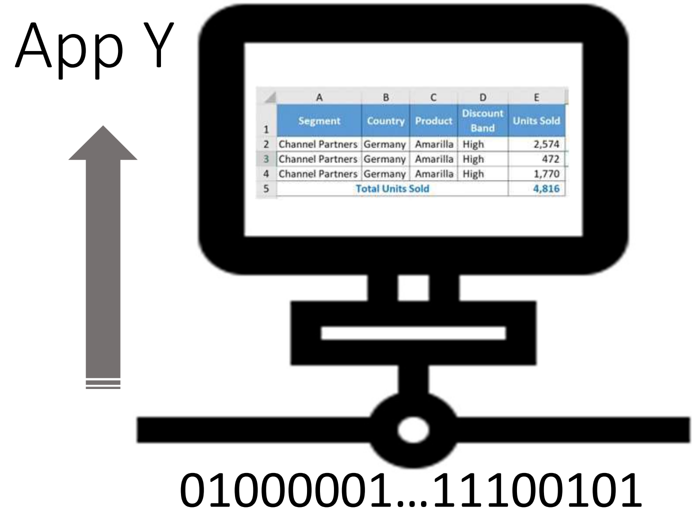
OSI Model (cont.)
Outgoing PDU Construction

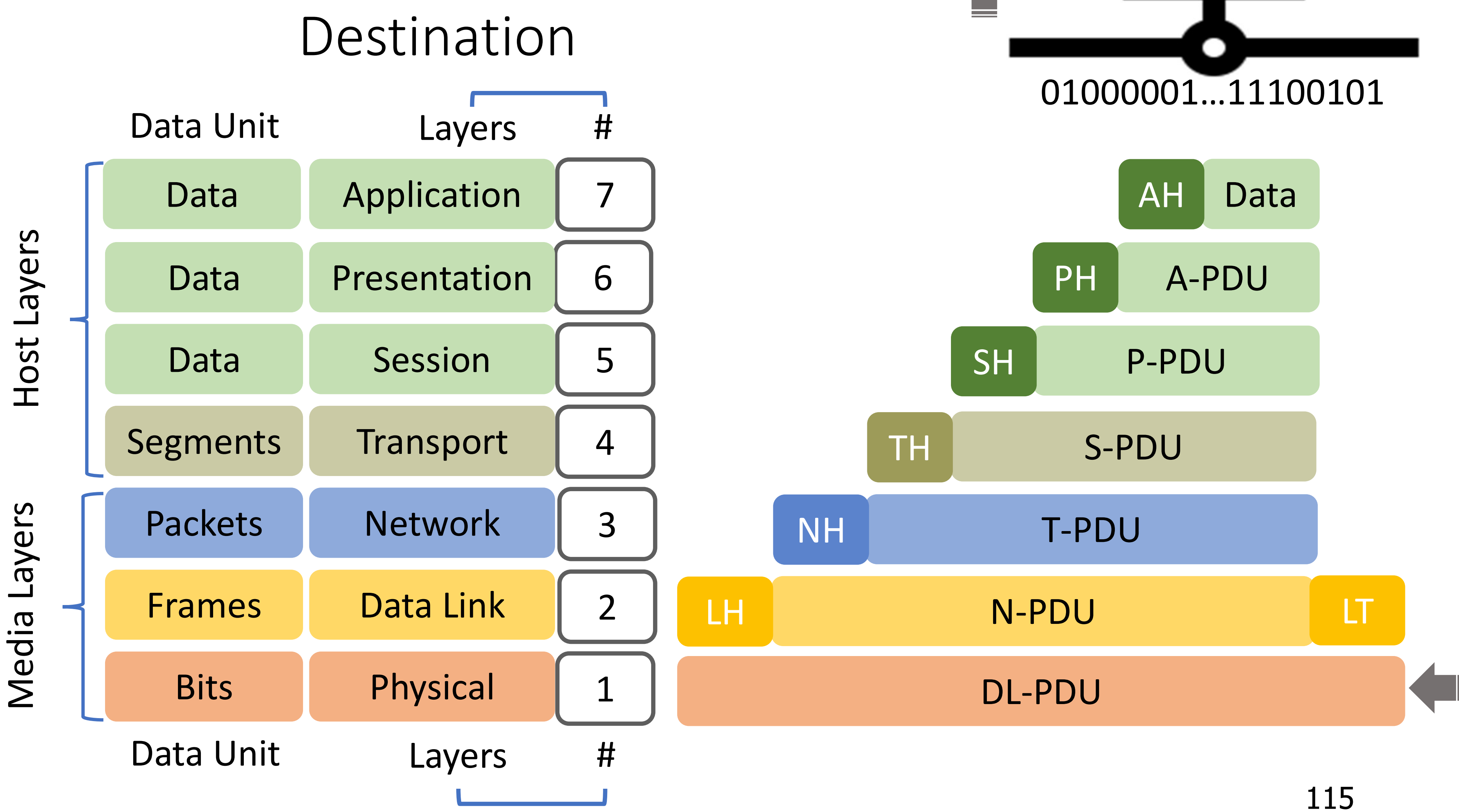




OSI Model (cont.)

Incoming PDU Reduction





OSI Layers

- Physical Layer (Layer 1)
  - o defines the means of transmitting raw bits rather than logical data packets over a physical data link connecting network devices (nodes)
- Data Link Layer (Layer 2)
  - o Means of activating, maintaining and deactivating a reliable link
  - o Error detection and control
  - o Higher layers may assume error free transmission

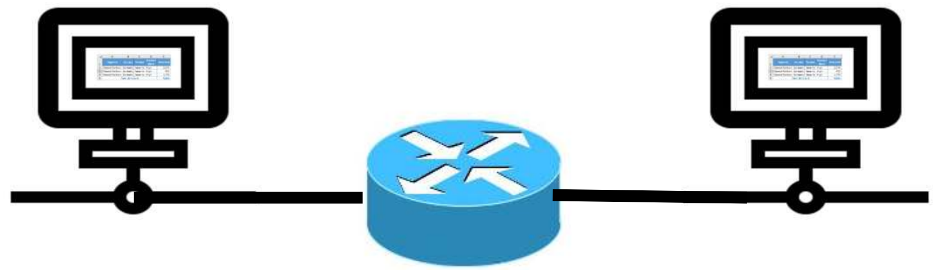
OSI Layers (cont.)

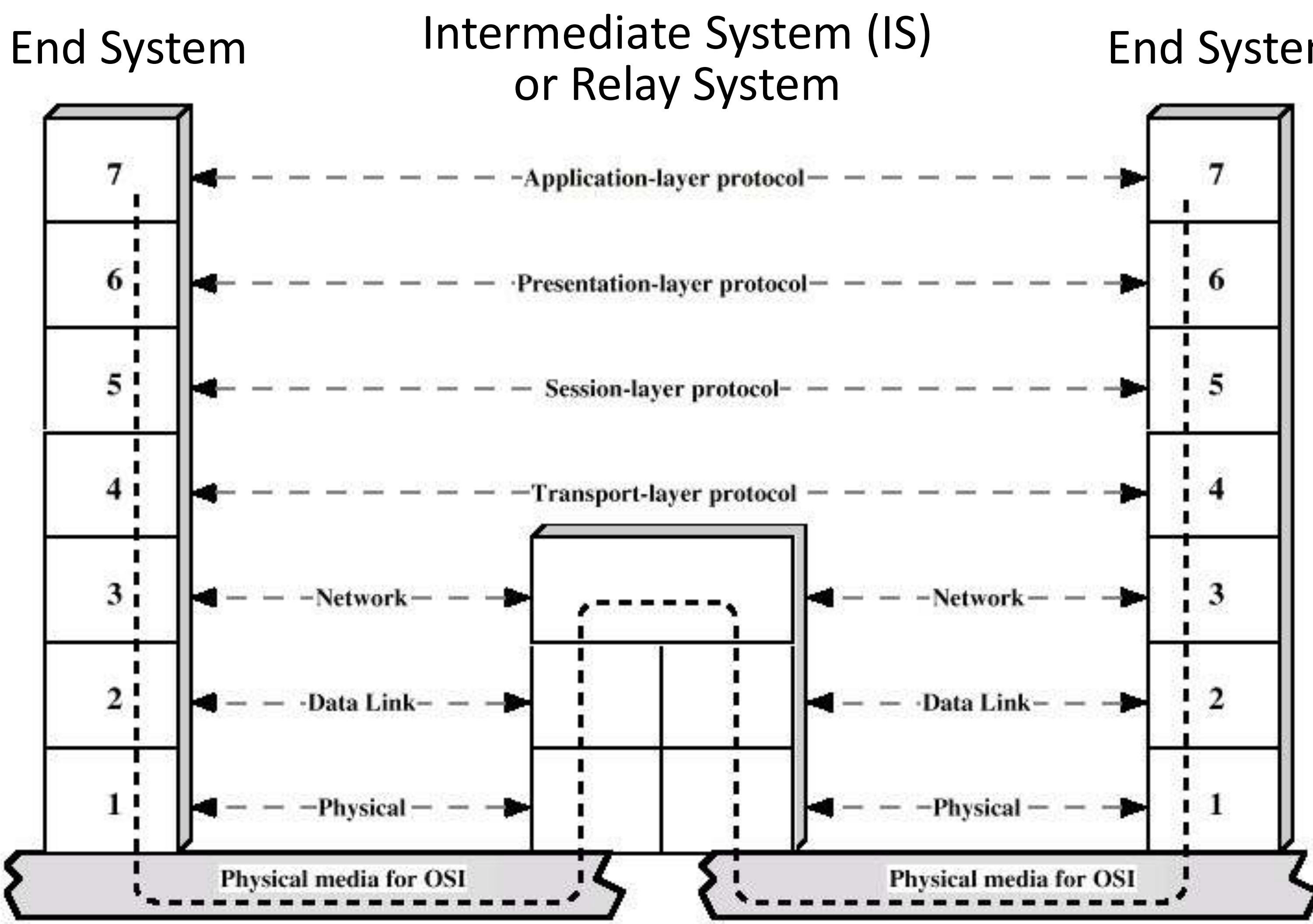
- Network Layer (Layer 3)
  - provides the means of transferring variable-length network packets from a source to a destination host via one or more networks
- Transport Layer (Layer 4)
  - Exchange of data between end systems
  - Error free
  - In sequence
  - No losses
  - No duplicates
  - Quality of service (QoS)

OSI Layers (cont.)

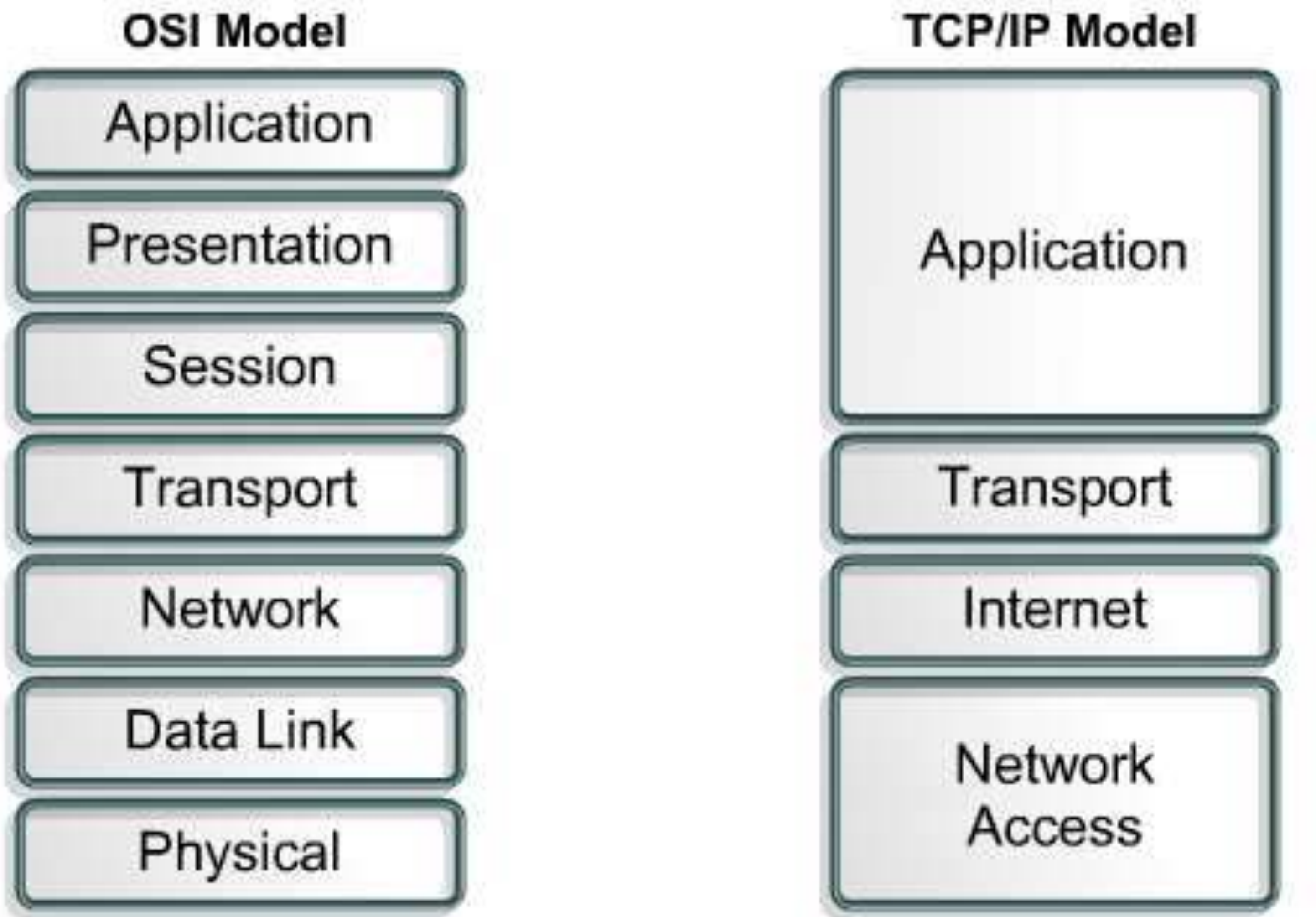
- Session Layer (Layer 5)
  - tracks the dialogues (sessions) between applications
  - controls the connections between multiple computers.
  - establishes, controls and ends the sessions between local and remote applications.
- Presentation Layer (Layer 6)
  - Data formats and coding
  - Data compression
  - Encryption
- Application Layer (Layer 7)
  - Means for applications to access OSI environment (user interface for displaying received information to the user)

Use of a Relay





TCP/IP Model


OSI Model
Application
Presentation
Session
Transport
Network
Data Link
Physical
TCP/IP Model
Application
Transport
Internet
Network
Access

TCP/IP Protocol Architecture

- Application Layer
  - o defines TCP/IP application protocols and how host programs interface with Transport layer services to use the network.
  - o includes all the higher-level protocols like DNS (Domain Name System), HTTP, Telnet, SSH (Secure Shell), FTP (File Transfer Protocol), SNMP (Simple Network Management Protocol), SMTP (Simple Mail Transfer Protocol), DHCP (Dynamic Host Configuration Protocol), etc.

TCP/IP Protocol Architecture (cont.)

- Transport Layer
  - permits devices on the source and destination hosts to carry on a conversation (end-to-end data transfer)
  - defines the level of service and status of the connection used when transporting data.
  - hides detail of underlying network
  - may include reliability mechanism (TCP)

The main protocols are TCP (Transmission Control Protocol) and UDP (User Datagram Protocol).

TCP/IP Protocol Architecture (cont.)

- Internet Layer
  - packs data into data packets (IP datagrams), which contain logical source and destination address (IP address) information that is used to forward the datagrams between hosts and across networks.
  - is also responsible for routing of IP datagrams.

The format of data that can be recognized by IP is called an IP datagram. It consists of two components, namely, the header and data, which need to be transmitted.

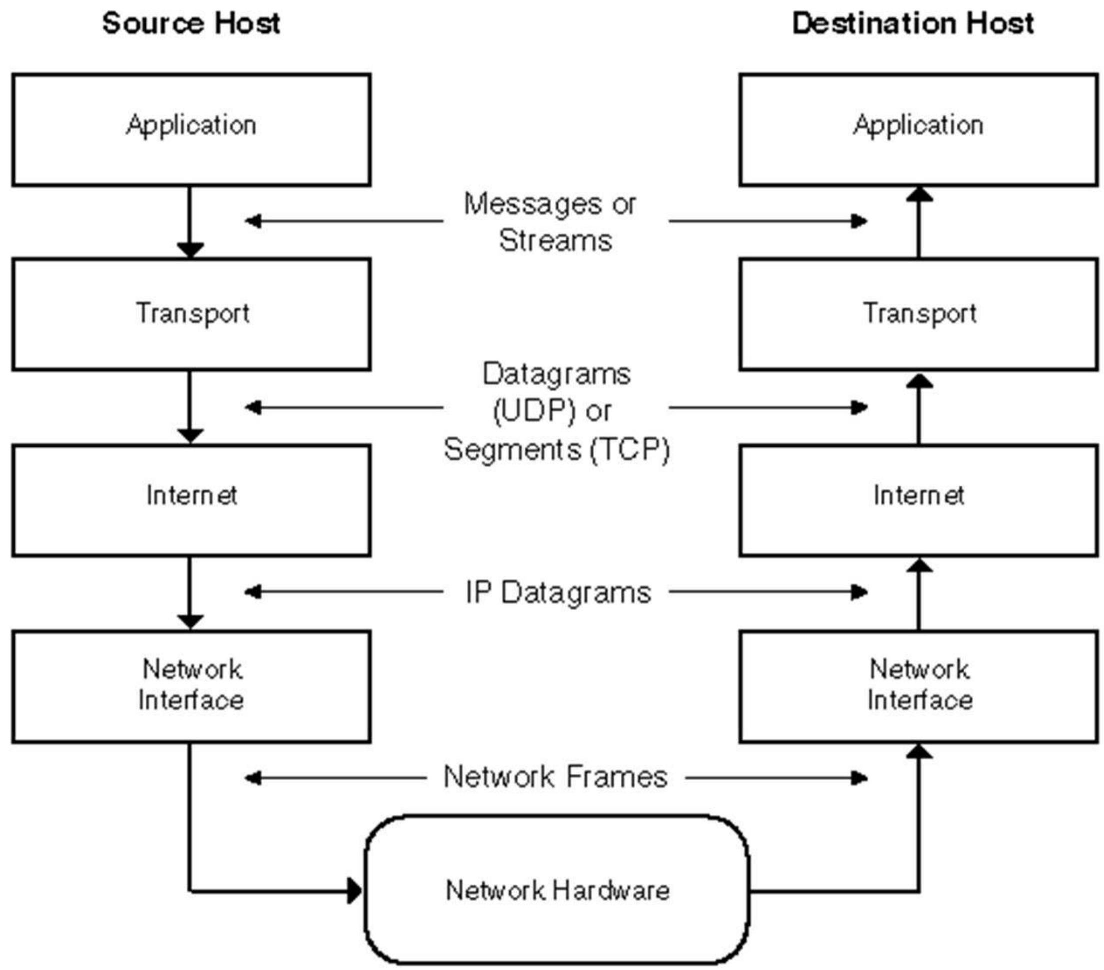
TCP/IP Protocol Architecture (cont.)

- Network Access Layer
  - defines how to use the network to transmit a frame, which is the data unit passed across the physical connection.
  - exchanges data between the computer and the physical network.
  - delivers data between two devices on the same network.

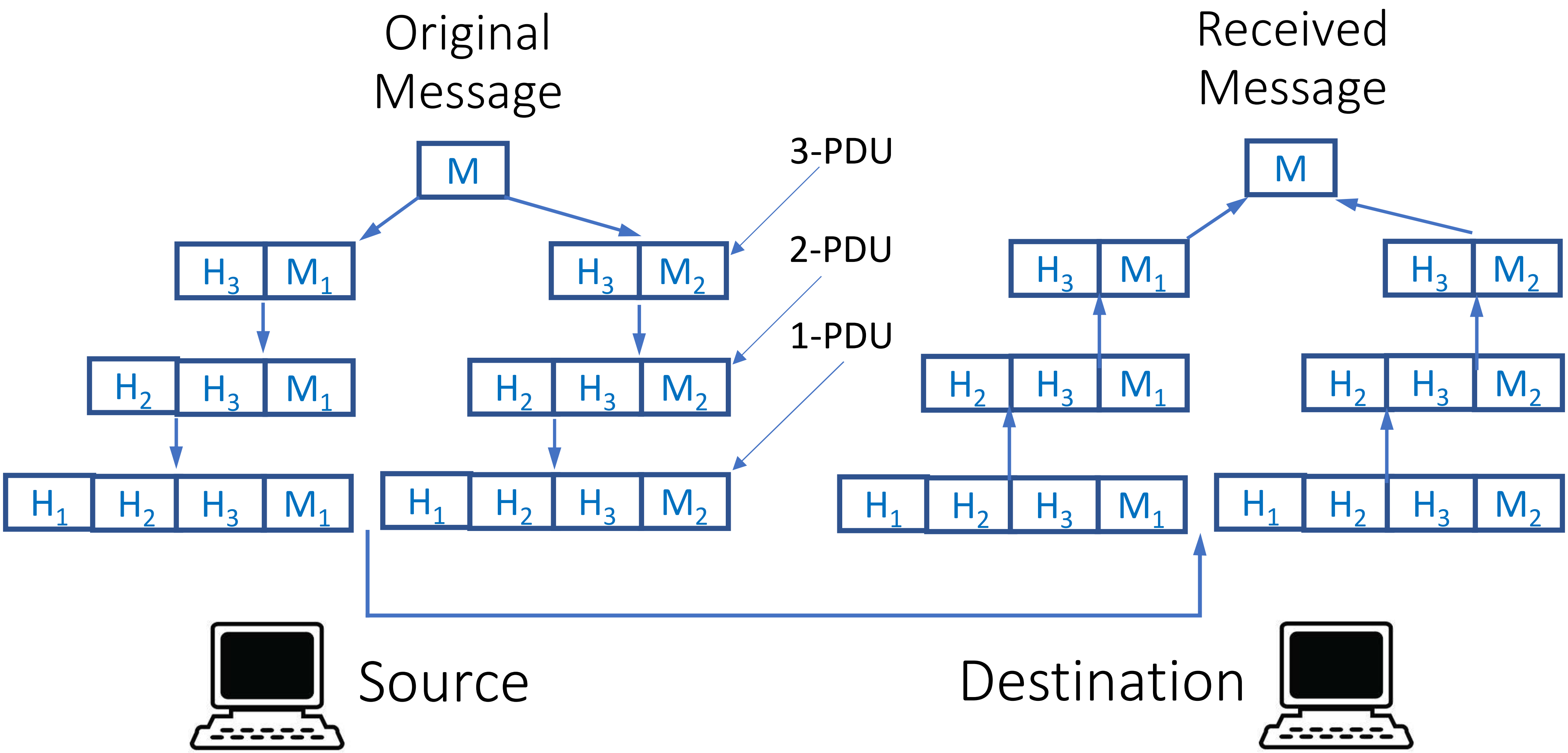
Unlike higher level protocols, the network access layer protocols must understand the details of the underlying physical network, such as the packet structure, maximum frame size, and the physical address scheme that is used.

TCP/IP: Peer-to-Peer Communication

- Process-to-Process Communication
- Host-to-Host Communication
- Source-to-Destination Communication
- Node-to-Node Communication

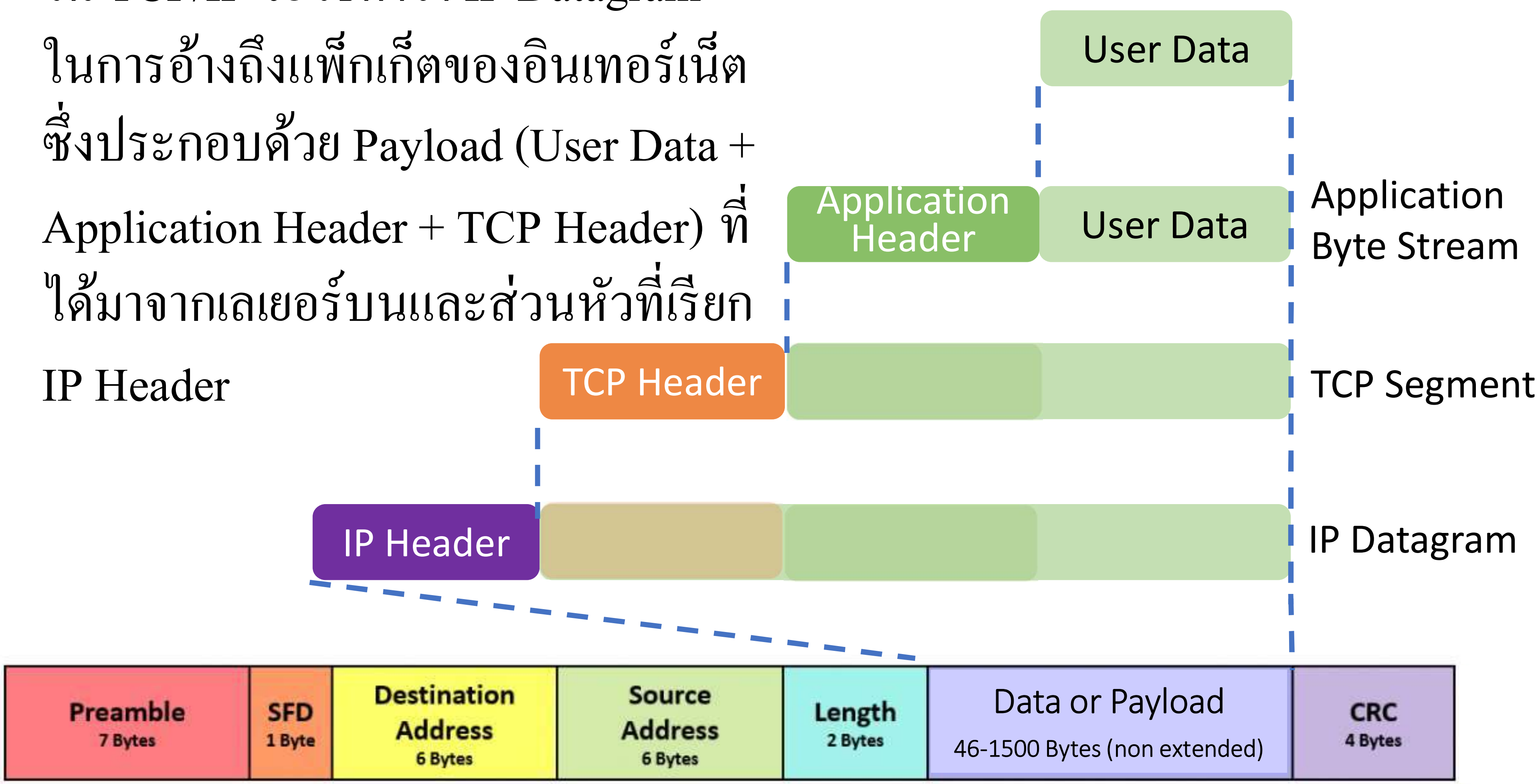


Different PDU's at Different Layers in the Protocol Architecture



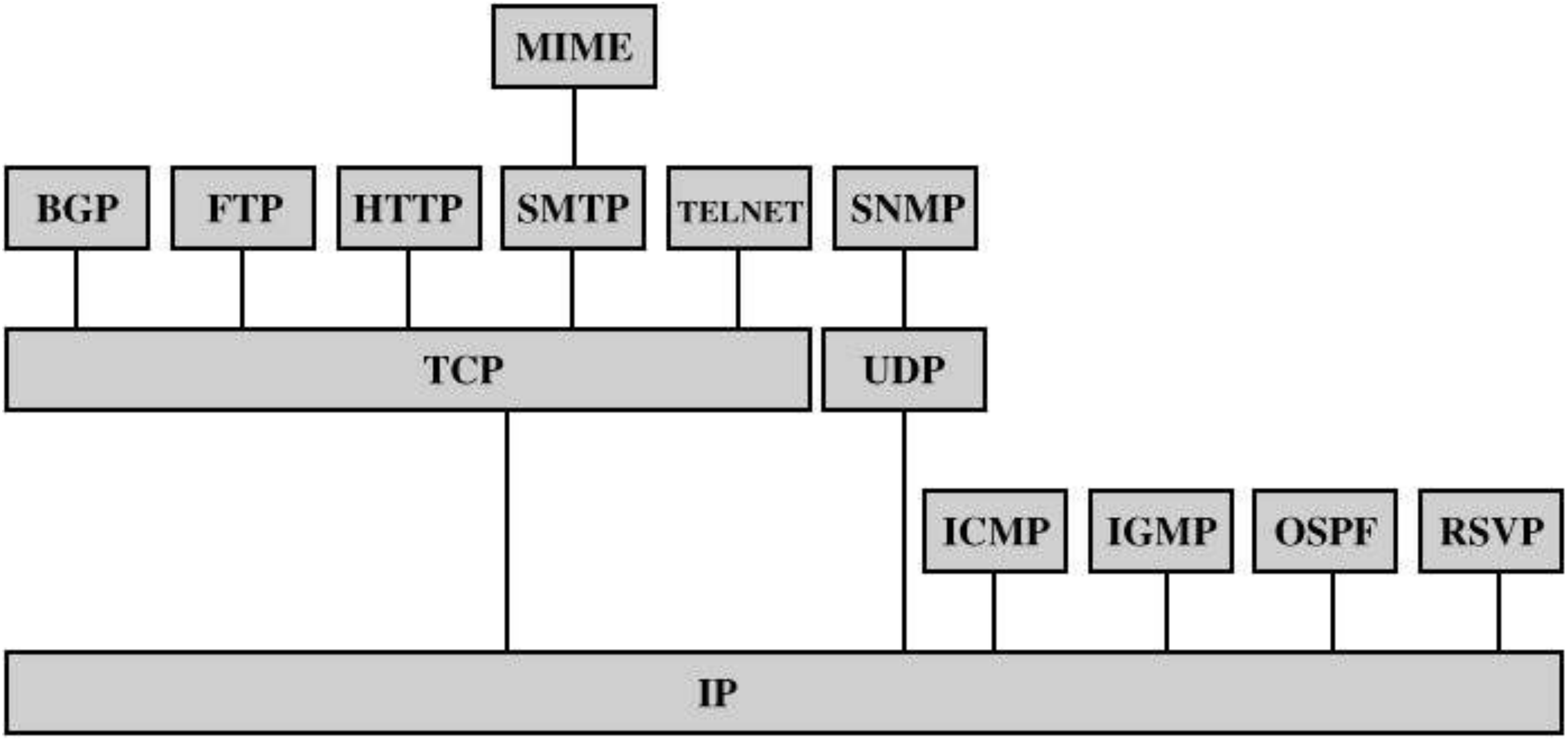
Protocol Data Units (PDUs) in TCP/IP

ใน TCP/IP จะใช้คำว่า IP Datagram ในการอ้างถึงแพ็กเก็ตของอินเทอร์เน็ต ซึ่งประกอบด้วย Payload (User Data + Application Header + TCP Header) ที่ได้มาจากเลเยอร์บนและส่วนหัวที่เรียก IP Header



Ethernet Frame Format

Some Protocols in TCP/IP Suite



BGP = Border Gateway Protocol
FTP = File Transfer Protocol
HTTP = Hypertext Transfer Protocol
ICMP = Internet Control Message Protocol
IGMP = Internet Group Management Protocol
IP = Internet Protocol
MIME = Multi-Purpose Internet Mail Extension
OSPF = Open Shortest Path First
RSVP = Resource ReSerVation Protocol
SMTP = Simple Mail Transfer Protocol
SNMP = Simple Network Management Protocol
TCP = Transmission Control Protocol
UDP = User Datagram Protocol

Transmission Control Protocol (TCP)

- In terms of the OSI model, TCP is a transport-layer protocol.
- It provides a reliable virtual-circuit connection between applications; that is, a connection is established before data transmission begins.
- Data is sent without errors or duplication and is received in the same order as it is sent.
- No boundaries are imposed on the data; TCP treats the data as a stream of bytes.

User Datagram Protocol (UDP)

- UDP is also a transport-layer protocol and is an alternative to TCP.
- It provides an unreliable datagram connection between applications.
- Data is transmitted link by link; there is no end-to-end connection.
- The service provides no guarantees.
- Data can be lost or duplicated, and datagrams can arrive out of order.

TCP vs. UDP

- TCP is a connection-oriented protocol, whereas UDP is a connectionless protocol.
- Overall, UDP is a much faster, simpler, and efficient protocol, however, retransmission of lost data packets is only possible with TCP.
- UDP is actually expected to work better than TCP in lossy networks (or congested networks).
- TCP is far better at transferring large quantities of data, but when the network fails it's more likely that UDP will get through.

Internet Protocol (IP)

- In terms of the OSI model, IP is a network-layer protocol.
- It provides a datagram service between applications, supporting both TCP and UDP.

Protocols and Architecture

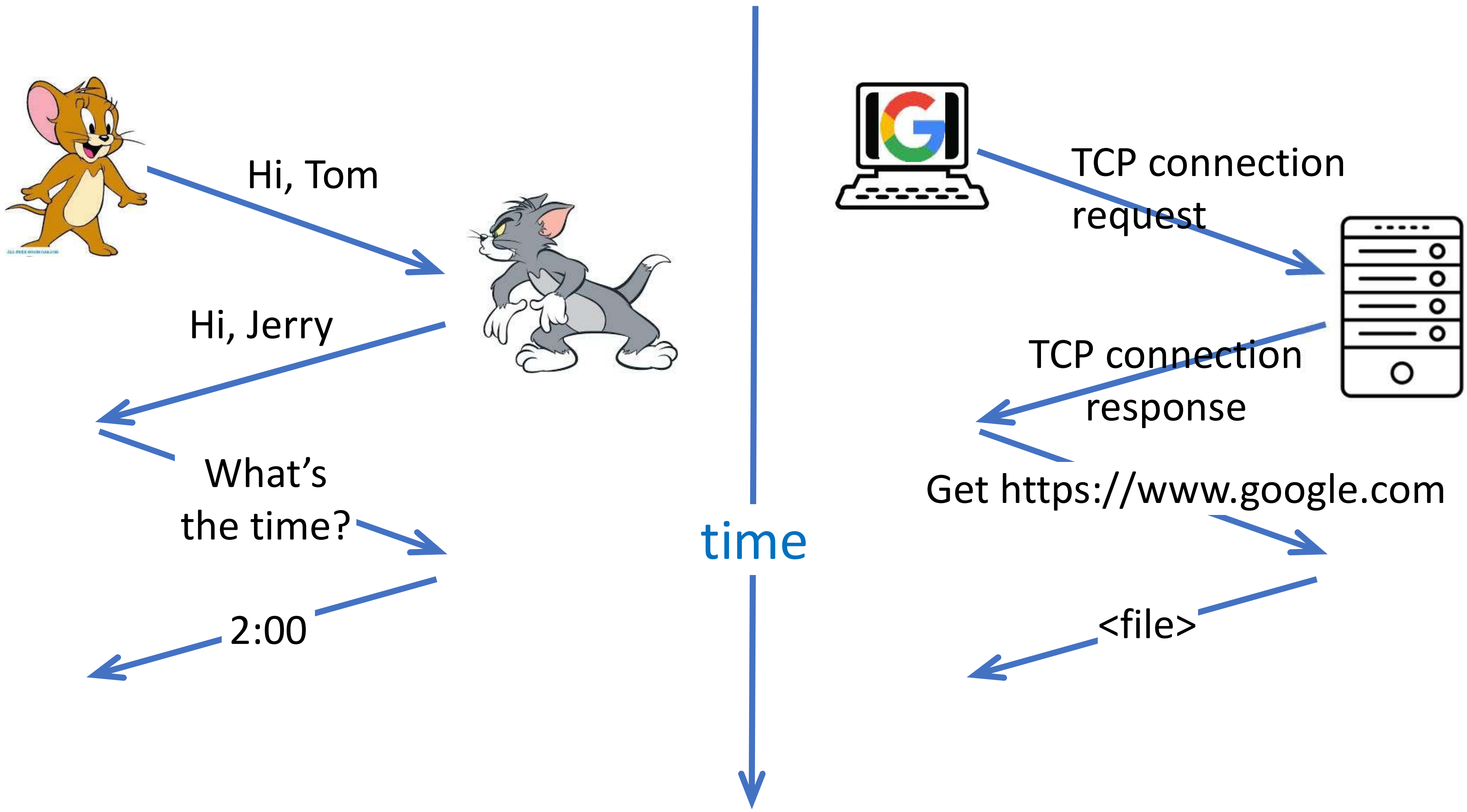
- Protocol and Protocol Suite
- Protocol Functions

Protocol and Protocol Suite
โพรโทคอลและชุดโพรโทคอล

- A **protocol** is a set of rules that govern how systems communicate.
- For networking they govern how data is transferred from one system to another.
- A **protocol suite** is a collection of protocols that are designed to work together.
- Most network protocol suites are structured as a series of layers, sometimes collectively referred to as a **protocol stack**.

What is a protocol?

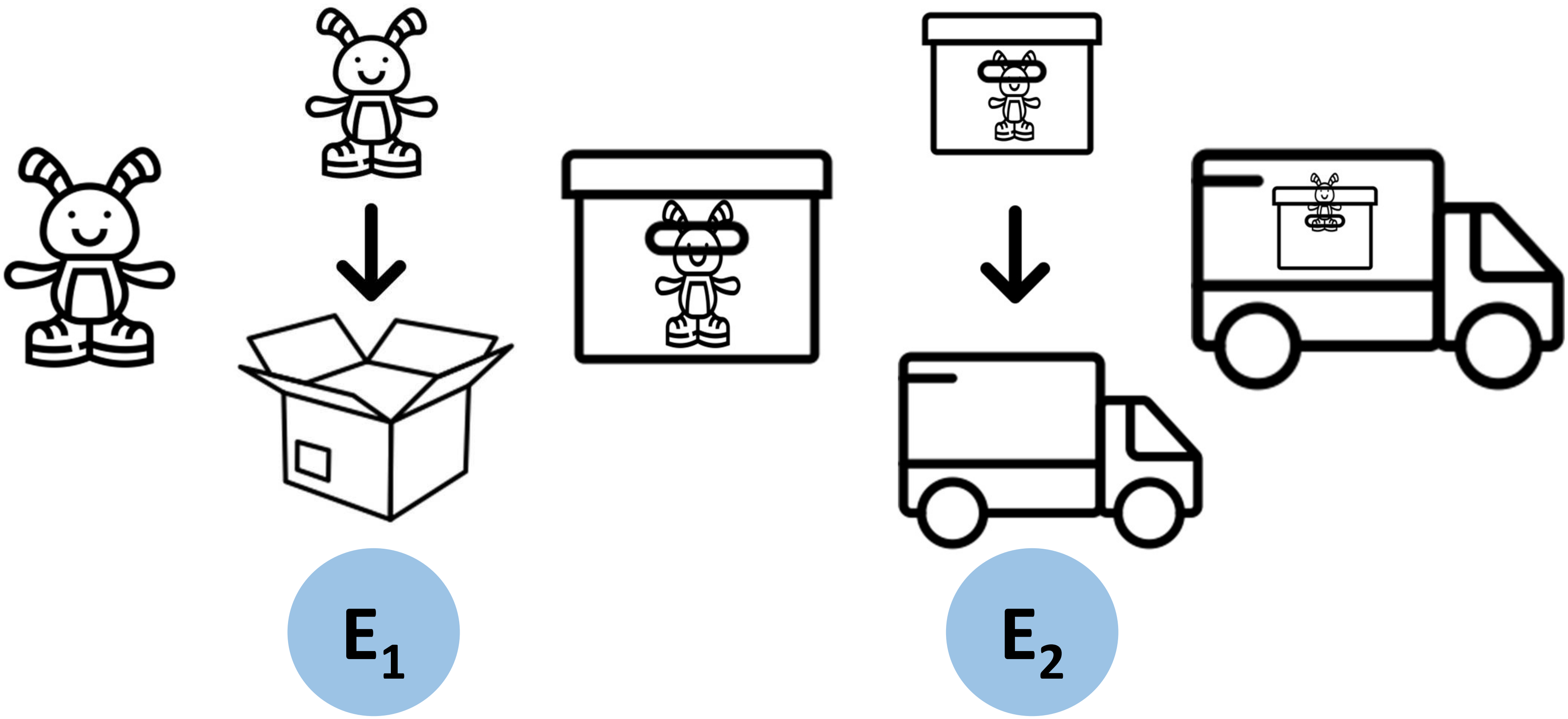
A Human Protocol vs A Computer Network Protocol



Protocol Functions

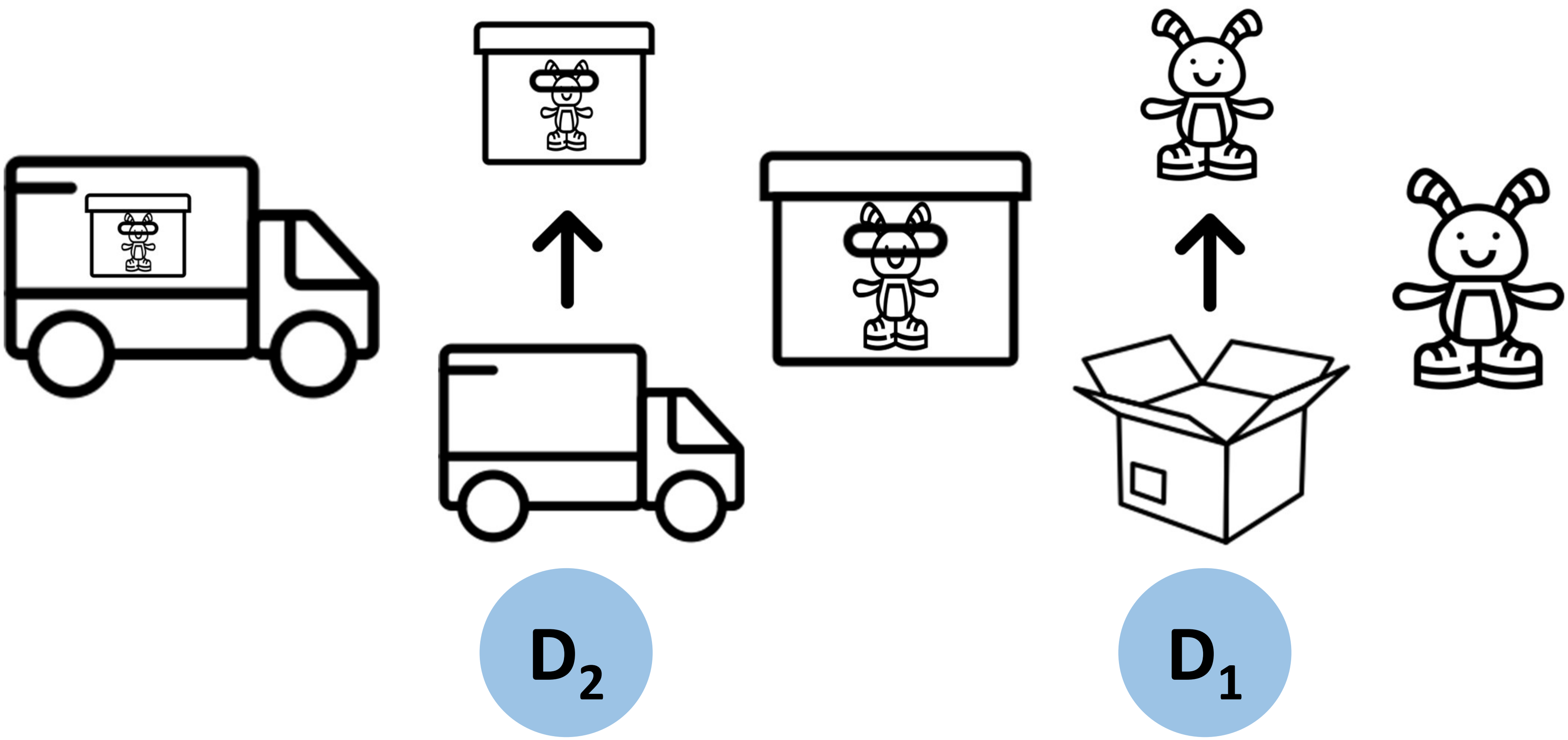
1. Encapsulation
2. Segmentation and reassembly (SAR)
3. Connection control
4. Ordered delivery
5. Flow control
6. Error control
7. Addressing
8. Multiplexing
9. Transmission services

1. Encapsulation (การห่อหุ้ม)



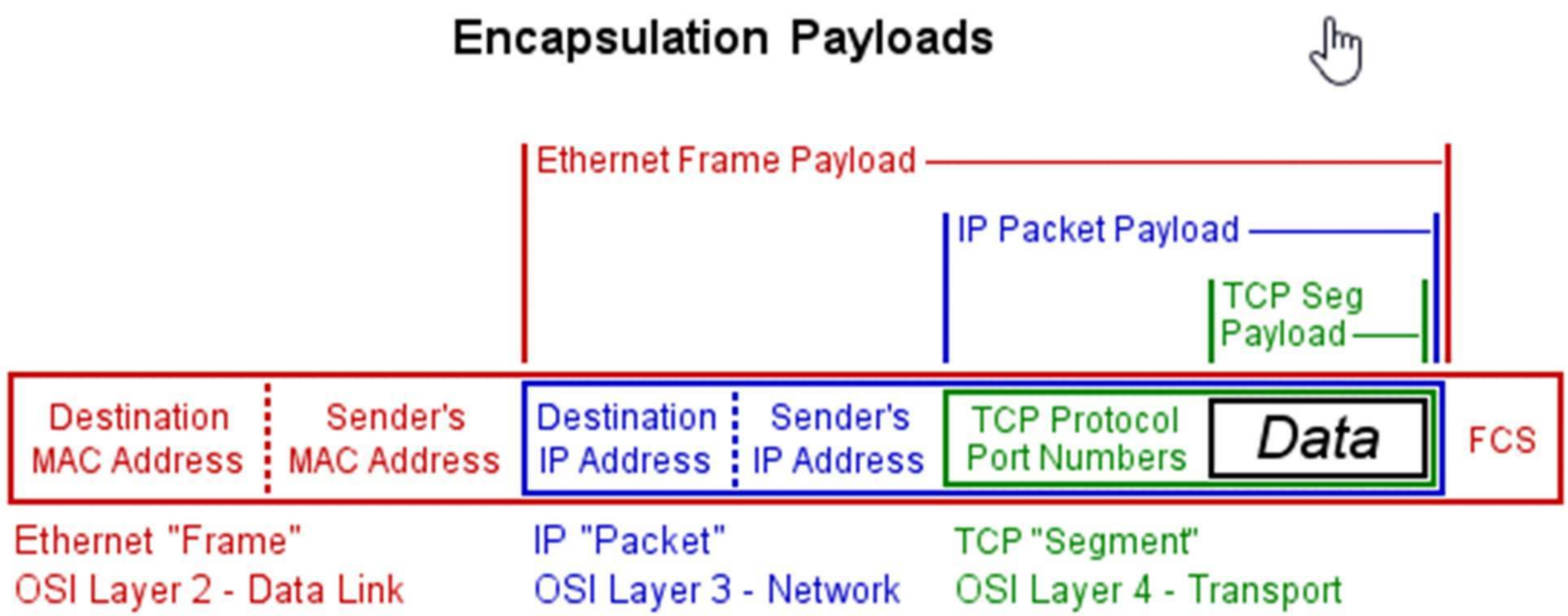
Decapsulation

การนำข้อมูลออกจากการทำ Encapsulation

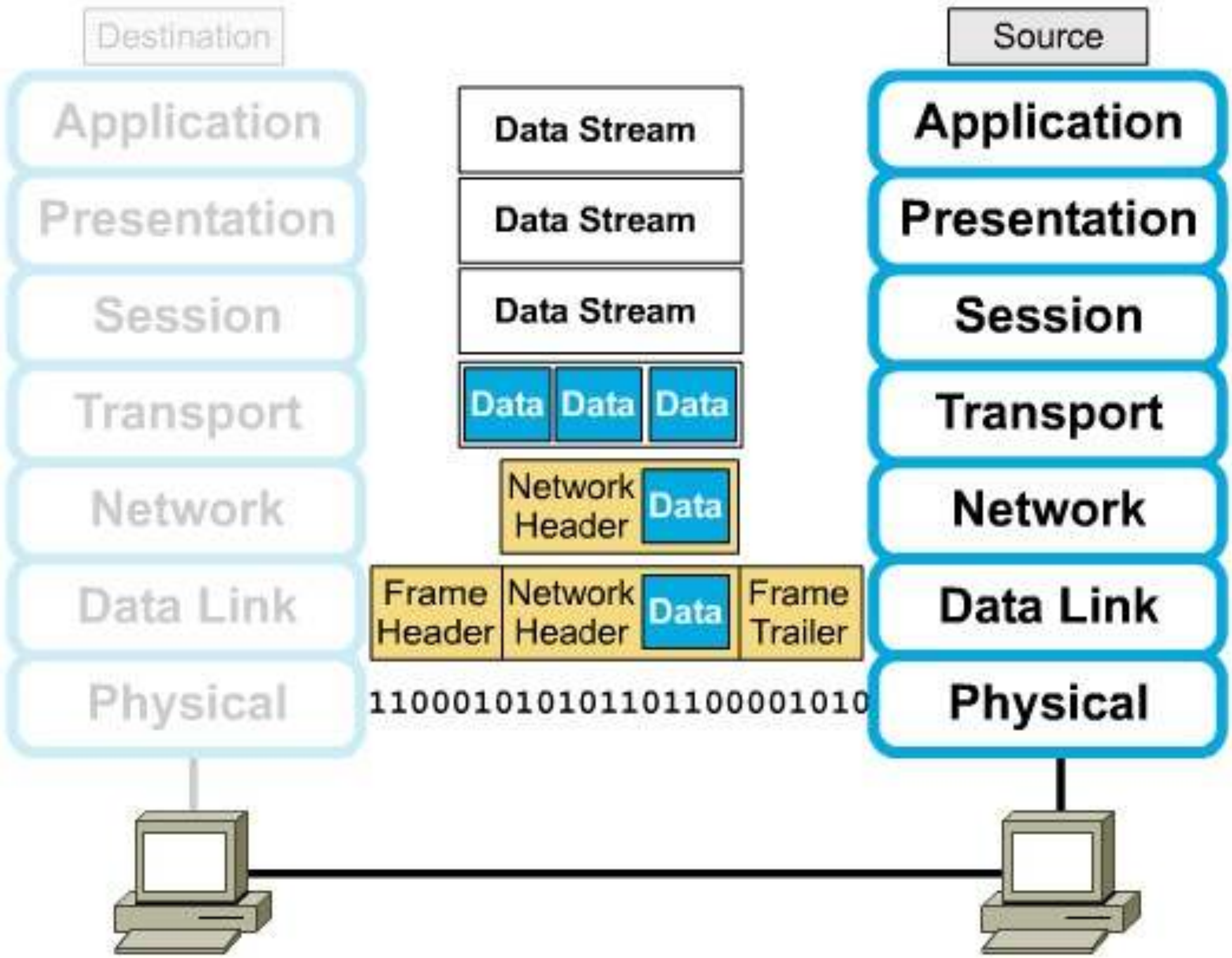


1. Encapsulation (cont.)

- Addition of control information to data
  - Addressing information
  - Error-detecting code, e.g. frame check sequence (FCS)
  - Protocol control



1. Encapsulation (cont.)



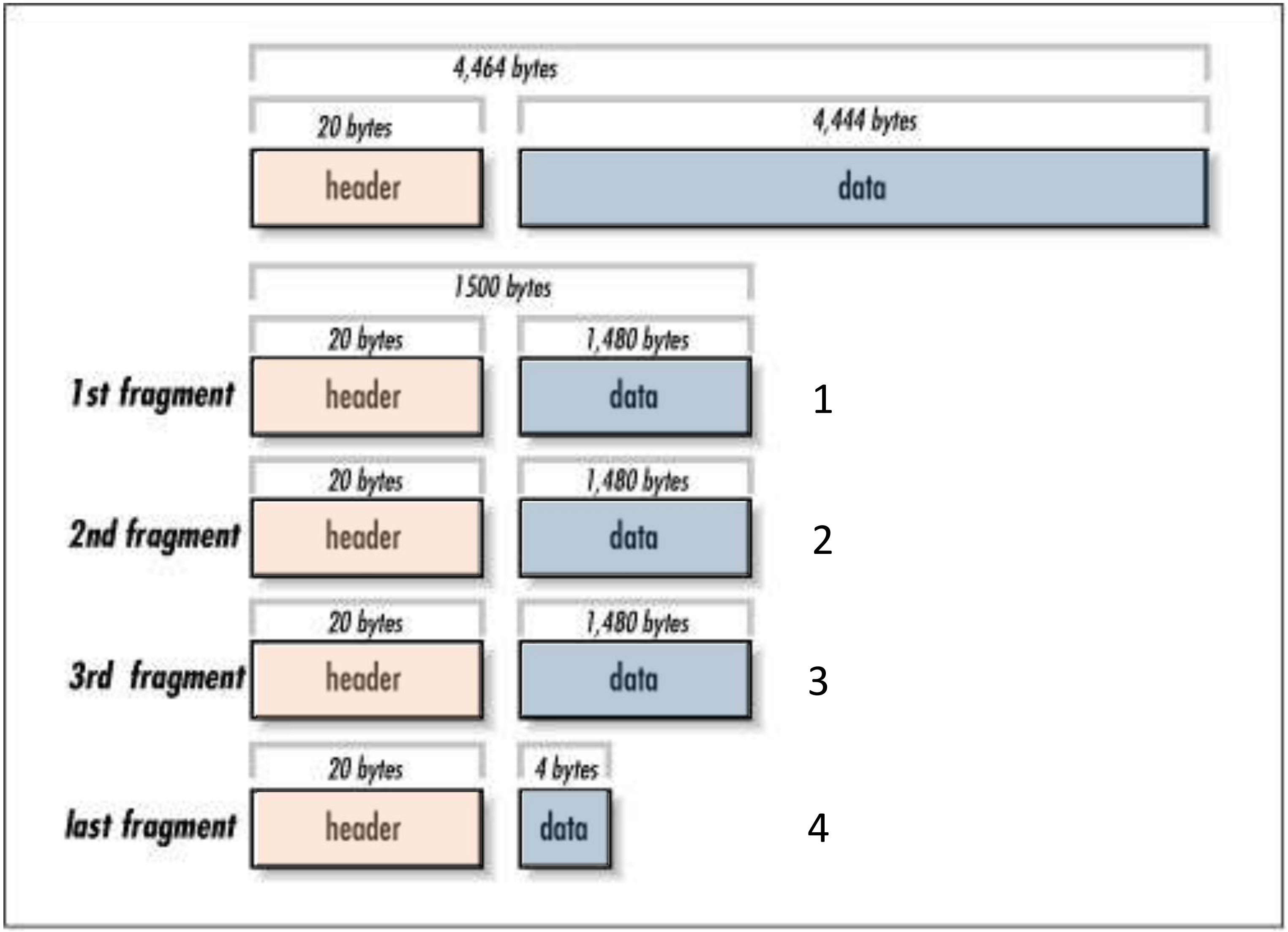
2. Segmentation and Reassembly

- In a packet-switched network, segmentation and reassembly (SAR; การแบ่งส่วนและการรวมส่วน) is a process of acquiring of incoming packets from upper layer and breaking them into smaller packets before transmission and reassembling them into the proper order at the receiving end of the communication.
- In TCP/IP, it is known as fragmentation.
- In the Open Systems Interconnection (OSI) model, SAR is performed in the Transport layer at both ends.

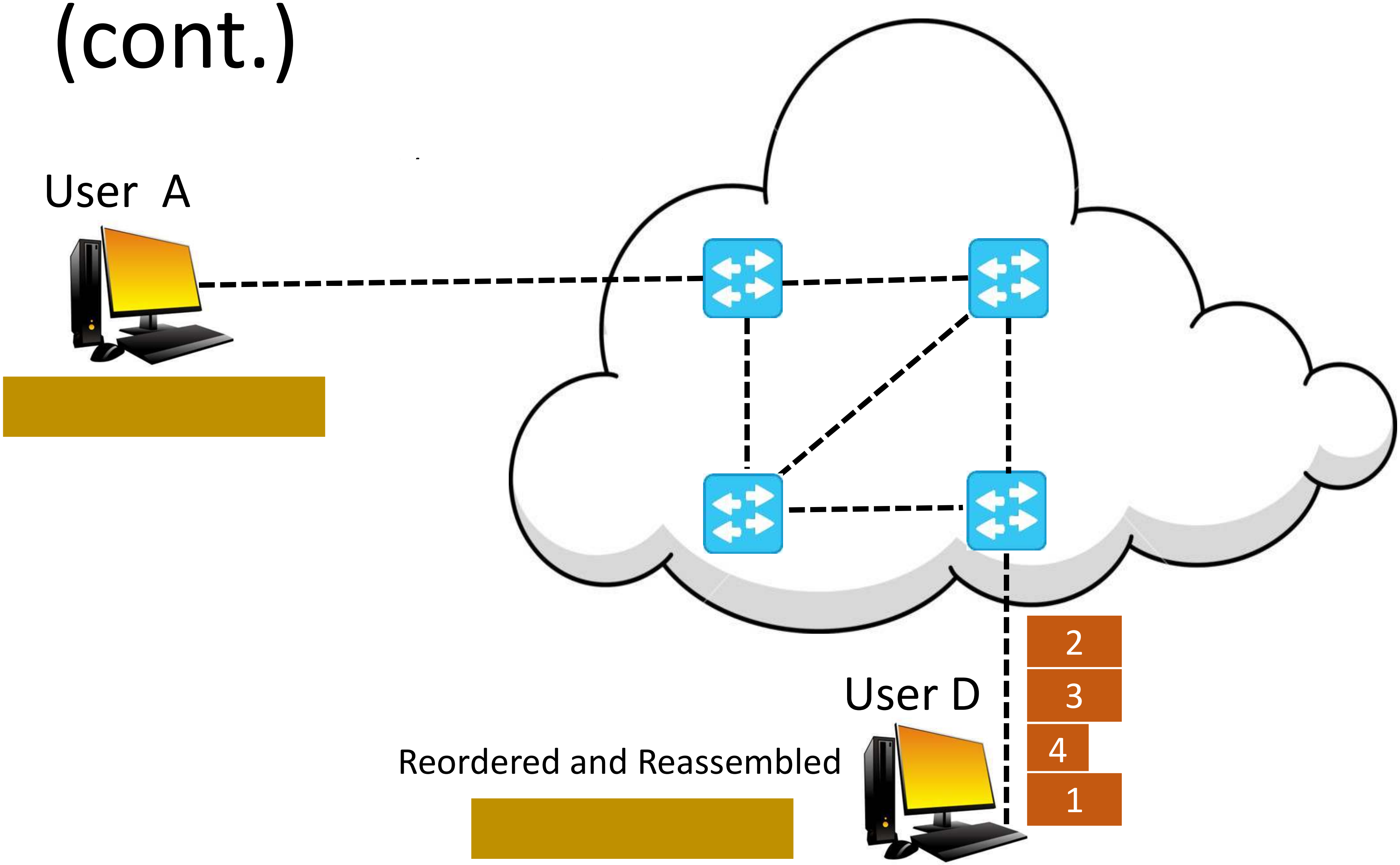
2. Segmentation and Reassembly (cont.)

- Maximum transmission unit (MTU) is the maximum length of protocol data unit (PDU) that can be transmitted by a protocol in one instance; the packets are segmented accordingly.
- Larger MTU is associated with reduced overhead. Smaller MTU values can reduce network delay.
- The MTU size for Ethernet is 1,500 bytes.

2. Segmentation and Reassembly (cont.)



2. Segmentation and Reassembly (cont.)



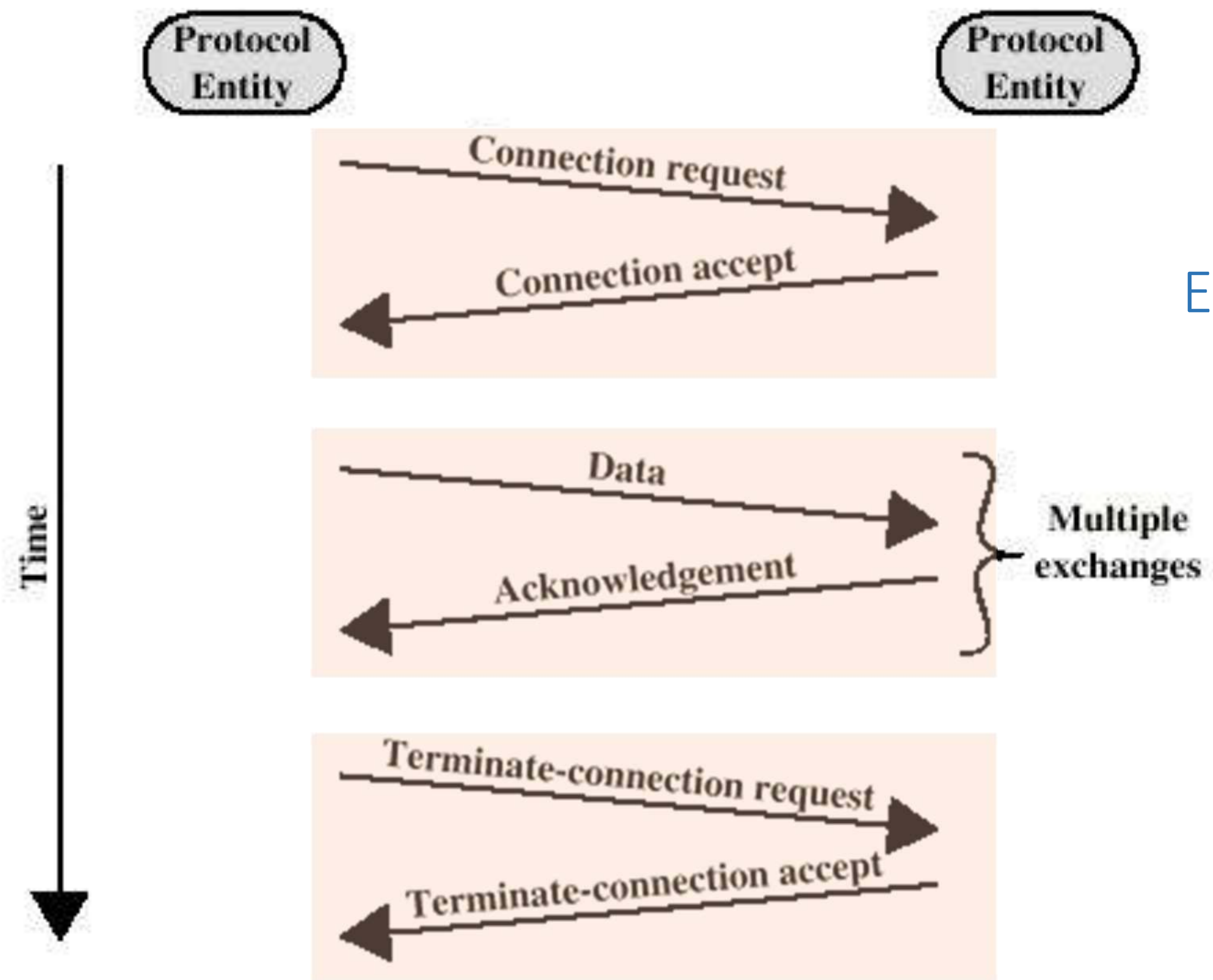
2. Segmentation and Reassembly (cont.)

- Advantages
  - More efficient error control
  - More equitable access to network facilities
  - Shorter delays
  - Smaller buffers needed
- Disadvantages
  - Overheads
  - Increased interrupts at receiver
  - More processing time

3. Connection Control

- Three phases occur for connection-oriented
  - Connection Establishment
  - Data transfer
  - Connection termination
- May be connection interruption and recovery
- To ensure connectivity, each TCP segment is numbered before transmission. During connection establishment each party uses a Random number generator to create initial sequence number (ISN), which is usually different in each direction. We know that a TCP sequence number is 32 bit.
- Sequence numbers are used for (4) Ordered delivery (5) Flow control and (6) Error control.

Connection Oriented Data Transfer



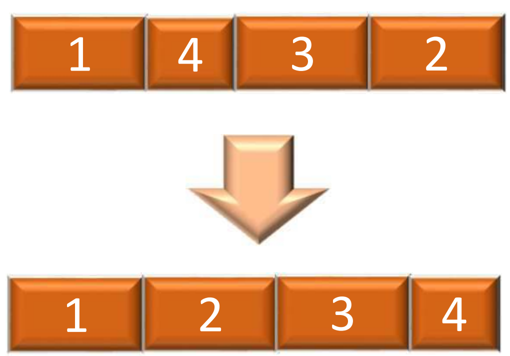
Connection Establishment

Data Transfer

Connection termination

4. Ordered Delivery

- PDUs may traverse different paths through network
- PDUs may arrive out of order
- Sequentially number PDUs to allow for ordering

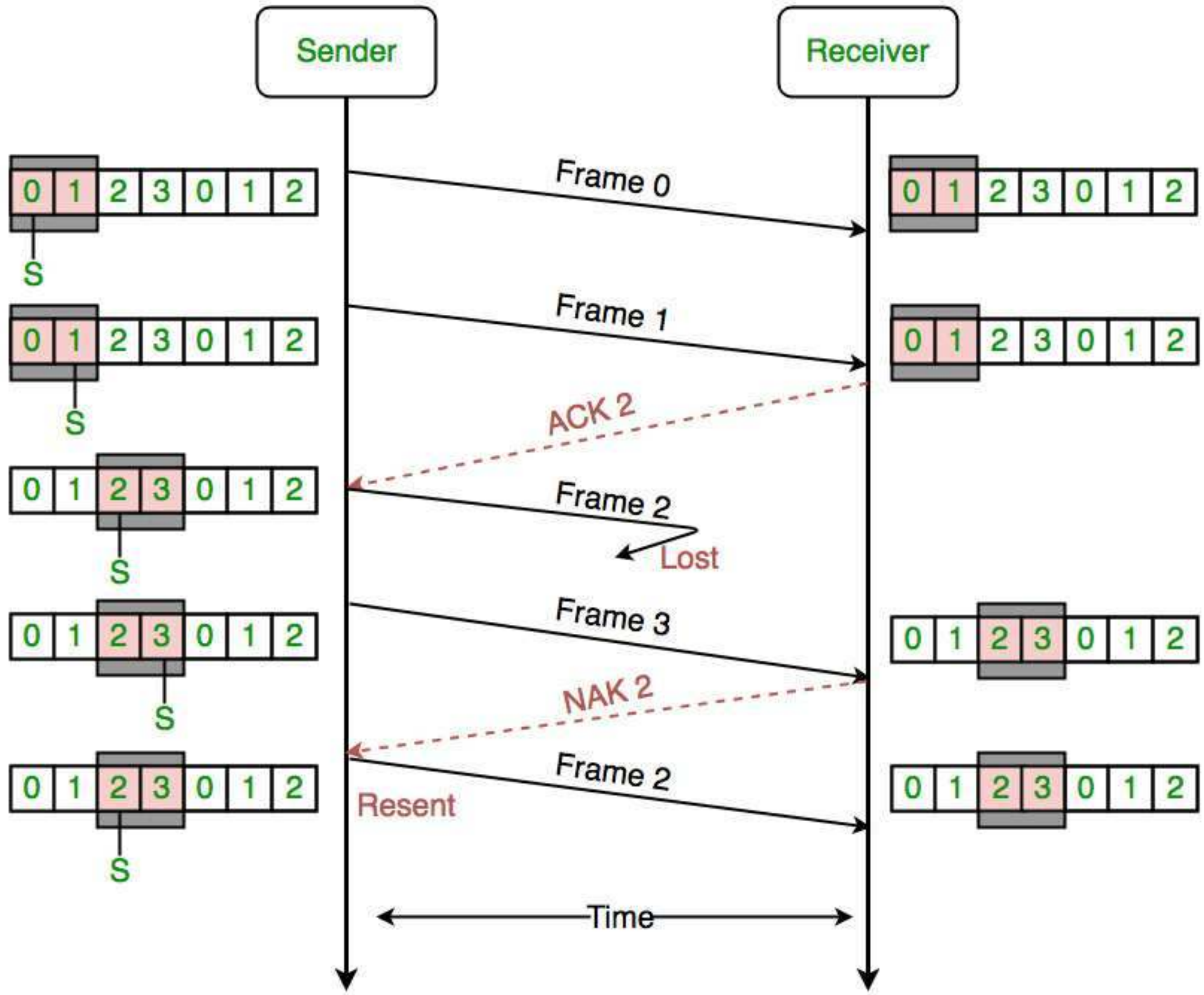


5. Flow Control

- Done by receiving entity
- Limit amount or rate of data
- Stop and Wait Protocol: the sender waits for an acknowledgment after transmitting each packet.
- Sliding Window Protocol: permits streamed communication by allowing the sender to transmit multiple packets before waiting for an acknowledgement.
- Needed at application as well as network layers

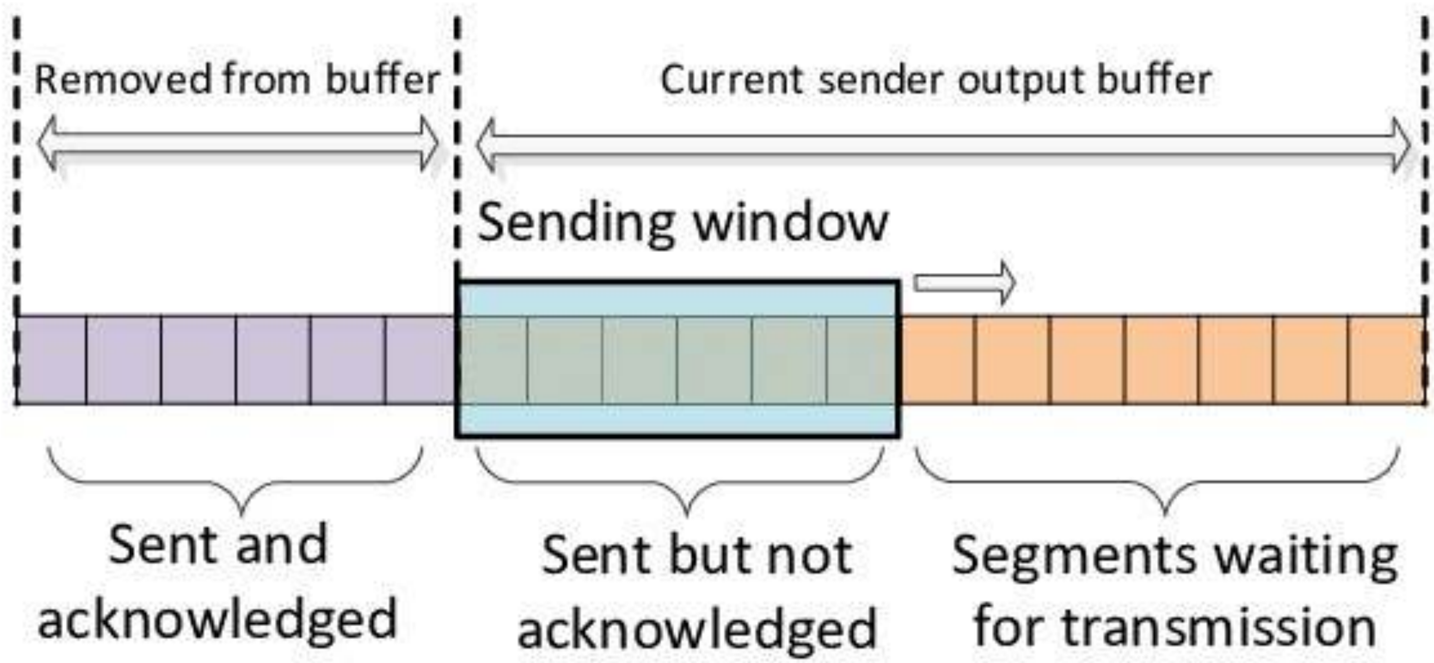
5. Flow Control (cont.)
Sliding Window Protocol

Sender Window Size (Ws=2) *Receiver Window Size (Wr=2)*



5. Flow Control (cont.)

Sliding Window Protocol



TCP Sliding Window Mechanism

6. Error Control

- Guard against loss or damage
- Error detection
  - Sender inserts error detecting bits
  - Receiver checks these bits
  - If OK, acknowledge
  - If error, discard packet
- Retransmission
  - If no acknowledge in given time, re-transmit
- Performed at various levels

7. Addressing

7.1 Addressing Level

7.2 Addressing Scope

7.3 Connection Identifiers

7.4 Addressing Mode

7.1 Addressing Level

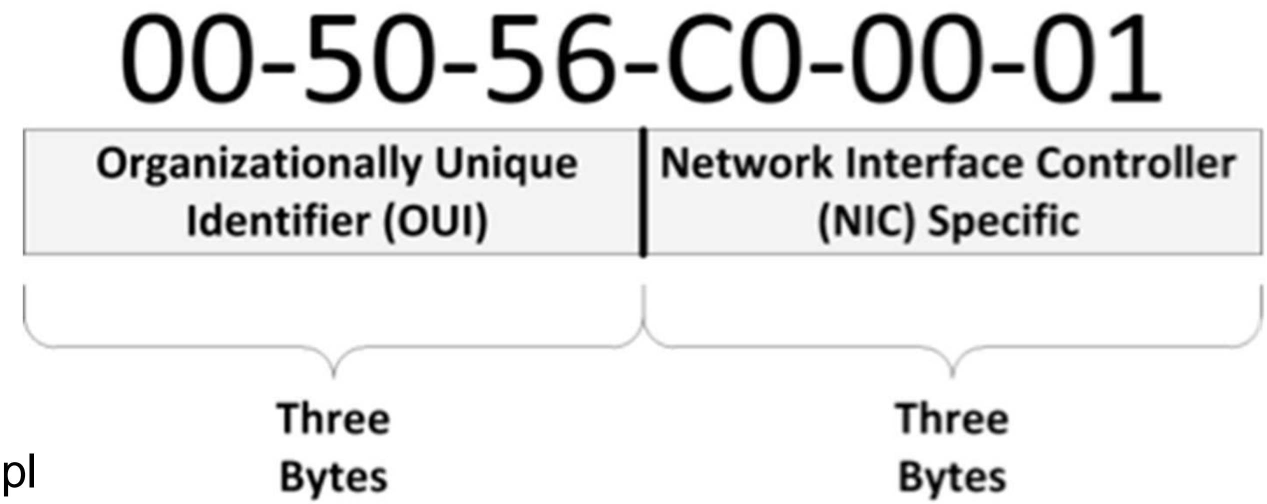
- Level in architecture at which entity is named
- Unique address for each end system (computer) and connecting device (router)
- Network level address
  - IP or Internet address (TCP/IP)
  - Network service access point or NSAP (OSI)
- Process within the system
  - Port number (TCP/IP)
  - Service access point or SAP (OSI)

7.2 Addressing Scope

- Global non-ambiguity
  - Global address identifies unique system
  - There is only one system with address X
- Global applicability
  - It is possible at any system (any address) to identify any other system (address) by the global address of the other system
  - Address X identifies that system from anywhere on the network
- e.g., MAC address on IEEE 802.x networks (LANs)

7.2 Addressing Scope – MAC Address

https://aruljohn.com/mac.pl

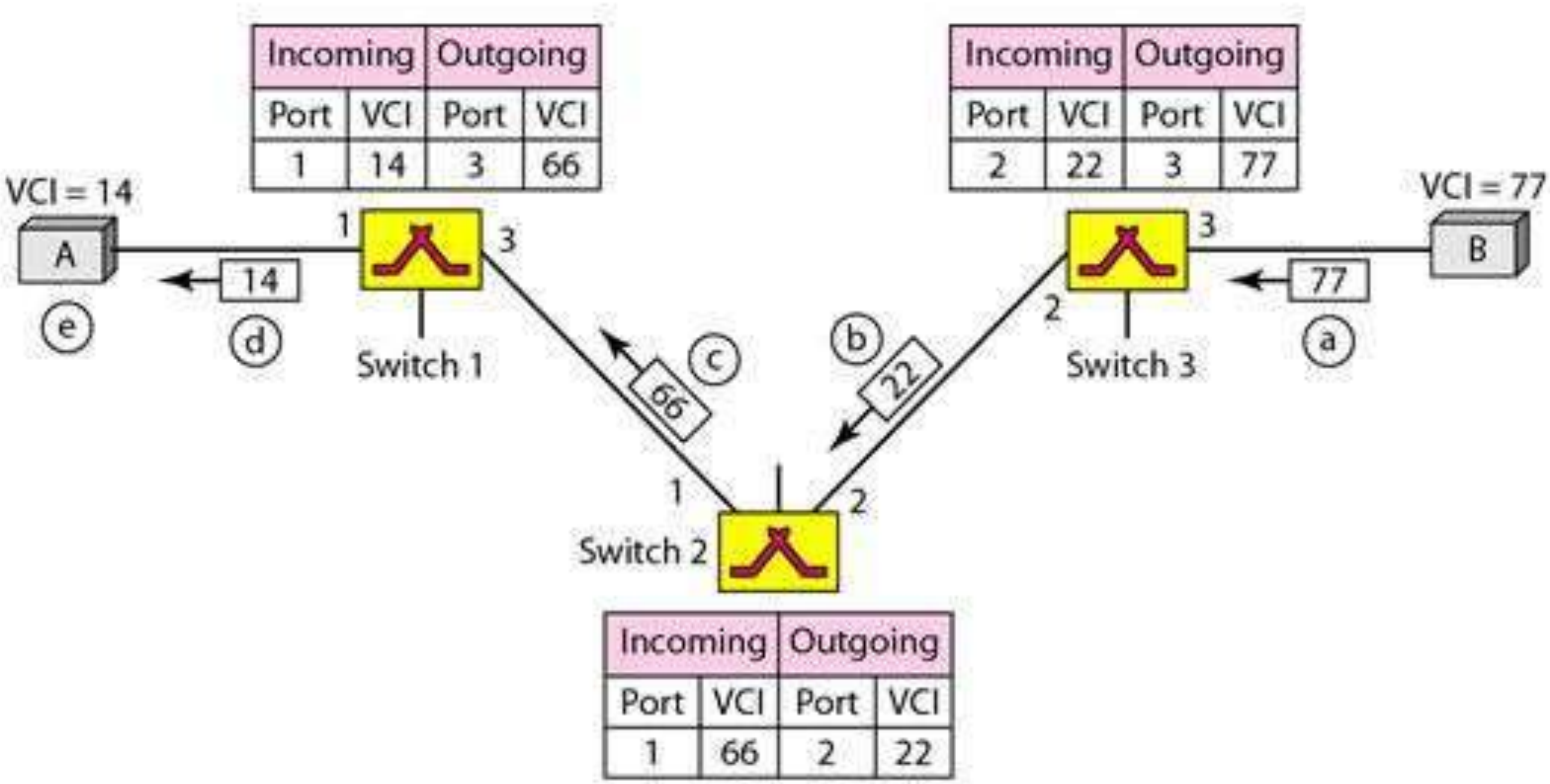


- Many network technologies, like Ethernet, use unique addresses to identify the network card, which is used to access the network.
- These addresses are usually called the Media Access Control address or MAC address. Each device is assigned a MAC address by its manufacturer.

7.3 Connection Identifiers

- Connection-oriented data transfer (virtual circuits)
- Allocate a connection name during the transfer phase
  - Reduced overhead as connection identifiers are shorter than global addresses
  - Routing may be fixed and identified by connection name
  - Entities may want multiple connections – multiplexing
  - Once connection established, end systems can maintain state information about connection

7.3 Connection Identifiers (cont.)

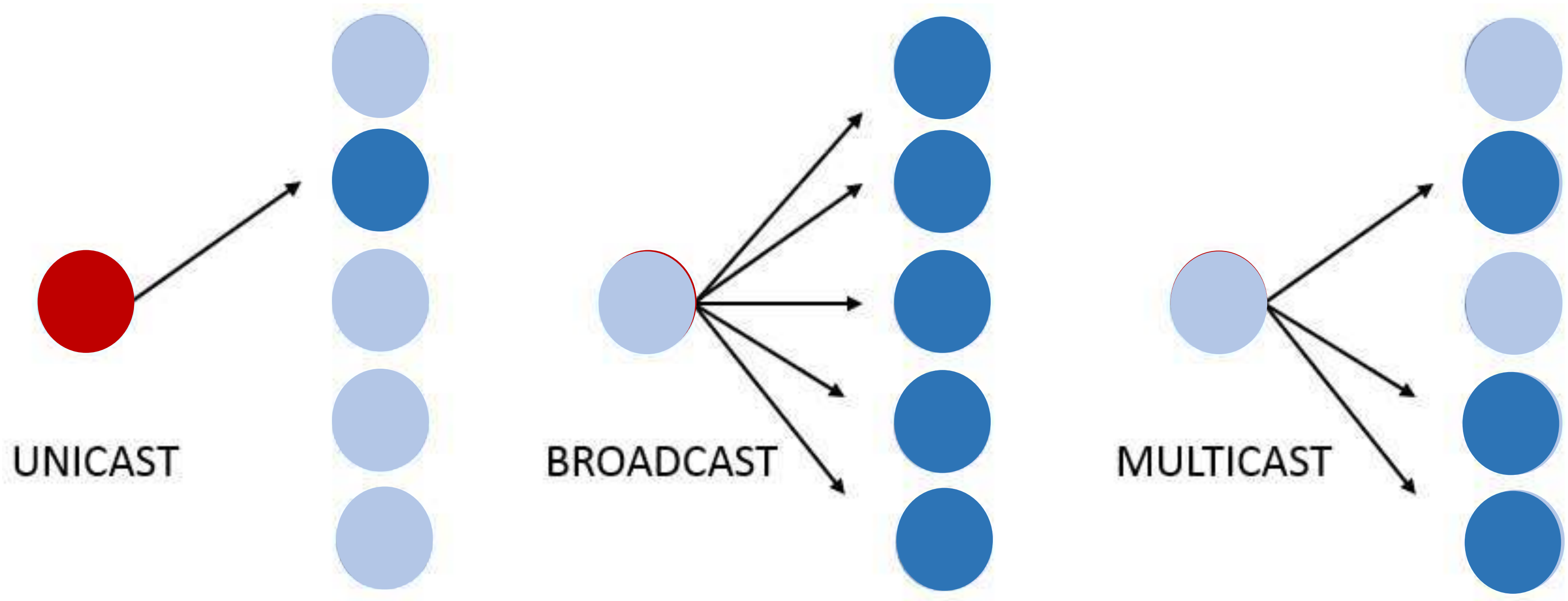


The figure show how the VCI in a data frame changes from one switch to another. Note that a VCI does not need to be a large number since each switch can use its own unique set of VCIs.

7.4 Addressing Mode

- Usually, an address refers to a single system
  - Unicast – Sent to one machine or person
- May address all entities within a domain
  - Broadcast – Sent to all machines or users
- May address a subset of the entities in a domain
  - Multicast – Sent to some machines or a group of users

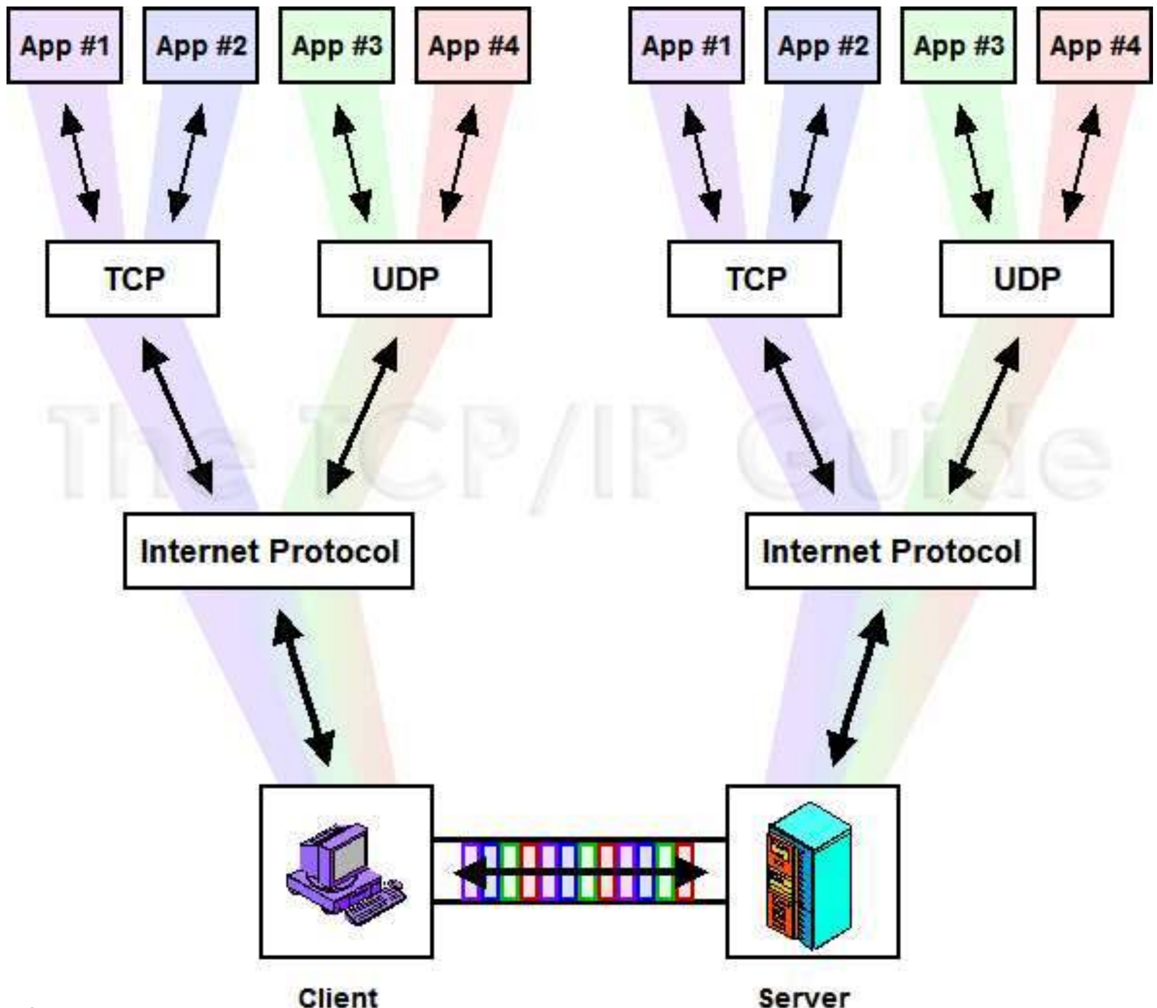
7.4 Addressing Mode (cont.)



8. Multiplexing

- Supporting multiple connections on one machine or sharing the link among multiple users
- In a typical machine running TCP/IP there are many different protocols and applications running simultaneously.
- This example shows four different applications communicating between a client and server machine.
- All four are multiplexed for transmission using the same IP software and physical connection; received data is demultiplexed and passed to the appropriate application.
- IP, TCP and UDP provide the means of keeping distinct the data from each application.

8. Multiplexing (cont.)
Multiplexing and Demultiplexing in TCP/IP



Credit: www.tcpipguide.com

8. Multiplexing (cont.)
Key Concept

- TCP/IP is designed to allow many different applications to send and receive data simultaneously using the same Internet Protocol software on a given device.
- To accomplish this, it is necessary to multiplex transmitted data from many sources as it is passed down to the IP layer.
- As a stream of IP datagrams is received, it is demultiplexed and the appropriate data passed to each application software instance on the receiving host.

9. Transmission Services

- Priority
  - e.g., control messages
- Quality of Service (QoS)
  - Minimum acceptable throughput
  - Maximum acceptable delay
- Security
  - Access restrictions

References

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Any Questions?